Supplementary Material to: Limits of Resolution in Quantum Machine Learning: A Statistical Noise Budget for Small Benchmarks

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S1 Additional Methods and Historical Experiments

The map of experiments is Table 2 of the main text.

Experiment identifiers. *Core*: the matched train-only QML experiment of the main text (15 seeds, same 2,273 fitting rows, all probabilities stored; Table S3). *Rerun*: the train-only classical 15-seed reruns of Random Forest, XGBoost and the LSTM. *Historical*: the canonical-pipeline experiments, whose per-seed AUCs were archived but whose test probabilities were stored only for seed 0 for the hybrids other than `LSTM+QNN`. The historical `LSTM+QNN` run likewise has stored test probabilities for seed 0 only; its 15-seed probabilities belong to the Core experiment. No paired comparisons are made for the three hybrids with seed-0 probabilities only. Historical results are descriptive and are not used for the inferential claims of the main text.

S1.1 Single-Seed Pairings

The main text compares the means of 15 seeds. For single runs, the paired standard error of two individual models is larger, and the verdict depends on which seeds are drawn.

Table S1: Single-seed pairings: observed 15-seed mean gaps and minimum detectable differences (MDE = 2.80× SE of the gap; $\alpha = 0.05$, power 0.80). Standard errors are averages over the 15×15 single-seed pairings; Table 4 gives the comparison of the 15-seed means. MDEs are computed from unrounded standard errors, so they can differ in the third decimal from 2.80× the SE shown. Last column: share of 15×15 single-seed pairings declared significant.

Pair	Observed gap	SE of gap, iid / block	MDE, iid / block	Declared significant, iid / block
RF − XGBoost	+0.021	0.013 / 0.014	**0.036 / 0.039**	27 % / 21 %
RF − LSTM	+0.015	0.014 / 0.010	**0.038 / 0.027**	3 % / 24 %
XGBoost − LSTM	−0.007	0.018 / 0.018	**0.052 / 0.051**	0 % / 0 %

Same-architecture control. DeLong tests between two seeds of the *same* model are significant in 1–4 % of pairs (iid) and 3–8 % (block). The classical models therefore rarely produce a nominally significant

difference from seeds alone. For the hybrids, whose seed variance is two to three times larger, an approximate version of this check is given in Section S3.4.

Same-architecture control and single-seed detection. Seed pairs of one model (e.g. seed i against seed j of Random Forest) are compared with the same two tests; the fraction declared significant is 1–4 % (DeLong) and 3–8 % (block). For cross-architecture pairs, all 15×15 combinations of seeds are compared. The same computation on the canonical pipeline gives 66 % (RF > XGBoost), 7 % (RF > LSTM) and 0 % (XGBoost versus LSTM), against 27 %, 3 % and 0 % for the train-only pipeline.

S1.2 Positive Control: Design

The MDE is a calculation; a positive control tests whether the benchmark behaves as it predicts. We take the test probabilities of Random Forest and add calibrated logit noise until the AUC falls by a known amount $\delta \in \{0.01, \ldots, 0.05\}$, with 15 seeds $\times$ 20 noise draws per δ ($\delta = 0$ is a null control in which every detection is a false positive). We then count how often the benchmark declares the undegraded model better. Two designs are used: the *same-seed* design, in which the degraded probabilities are derived from the same run (the loss δ is imposed exactly on the test set), and the *different-seed* design, which is the realistic one because the compared models come from different runs. Both are conditional perturbation experiments on the observed test set, not simulations from a known population: they check the detection machinery under one perturbation mechanism and do not establish coverage or power over new samples.

Positive control. Random Forest test probabilities p_i are transformed to $\sigma(\text{logit}(p_i) + \varepsilon_i)$ with Gaussian ε_i, calibrating the noise scale so that the AUC falls by δ; 15 seeds $\times$ 20 noise draws are used per δ. The same-seed design compares the degraded probabilities with those of the originating run; the different-seed design compares them with another seed of Random Forest. Both tests are applied to each pair and the proportion of pairs in which the undegraded model is declared significantly better (at $\alpha = 0.05$) is reported.

S1.3 Positive Control: Results

Table S2 and Figure S1 show how often the benchmark detects a Random Forest degraded by a known AUC loss. The 80 % detection point is reached at $\delta \approx 0.039$–0.043, consistent with the analytic MDE of the RF–XGBoost comparison. An imposed difference of 0.02 AUC, about the RF–XGBoost gap, is detected in only 17–30 % of runs. At $\delta = 0$ the false-detection rate is 2–5 %, and for $\delta \geq 0.01$ no significant result had the wrong sign.

Table S2: Positive control: percentage of runs in which the benchmark detects the undegraded Random Forest as better, as a function of the imposed AUC reduction δ (15 seeds $\times$ 20 noise draws per δ). Entries are iid (DeLong) / block bootstrap. At $\delta = 0$ every detection is a false positive.

δ **(imposed AUC reduction)**	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**
Same seed	—	8 / 15	20 / 30	52 / 62	84 / 80	97 / 92
Different seeds	2 / 5	7 / 12	17 / 27	47 / 57	75 / 76	93 / 92

The same-seed design fixes δ exactly on the test set; the different-seed design is the realistic one and is slightly less powerful below $\delta = 0.04$.

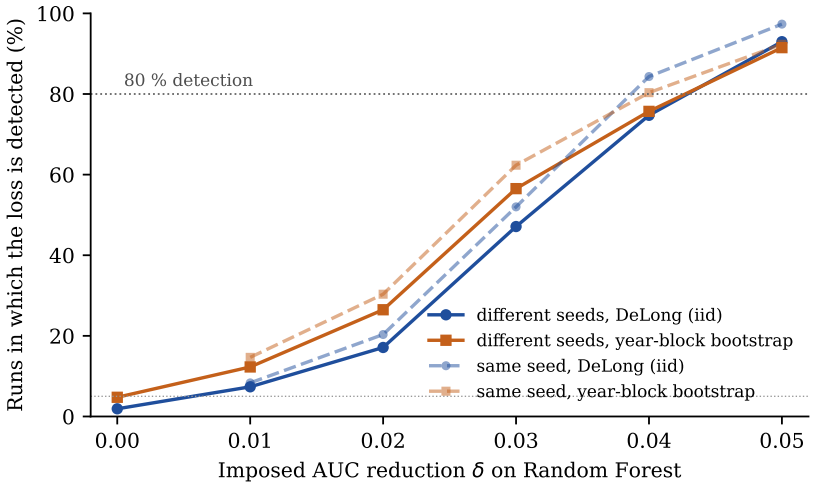


Figure S1: Detection rate of a known AUC loss imposed on Random Forest. Dashed lines: same-seed design; solid lines: different seeds (the realistic design). The detection rate reaches 80 % at $\delta \approx 0.04$.

Table S3: Specification of the matched 15-seed QML core experiment (script `qml_core_trainonly.py`; outputs `results_qml_core_trainonly.json` and `results_qml_core_trainonly_probs.npz`, storing 528 test probabilities for every seed).

Item	**Specification**
Pipeline	Train-only: anomalies standardized with years up to 1996 only; threshold P_{60} = 28.1 mm from the training partition; 35 input features.
Rows	Fit: 2,273 training windows (1938–1996); validation: 514 windows (1997–2007), used only for early stopping; test: 528 windows (2008–2026).
Seeds	0–14 for every stochastic model (`torch.manual_seed` and `numpy.random.seed`; `random_state` for Random Forest). The seed index is nominal across frameworks.
Random Forest	300 trees, maximum depth 8, balanced class weights, mean imputation fitted on the 2,273 rows.
Standalone LSTM	2-layer bidirectional LSTM (16 units per direction, dropout 0.3) over the first 12 input columns; its final states are concatenated with all input features, batch normalization, a 16-unit GELU layer (dropout 0.2, layer normalization) and a linear output.
Linear-4D control	2-layer bidirectional LSTM (8 units per direction, dropout 0.2) over the first 12 input columns; projection of the LSTM states and features to 4 dimensions (dropout 0.1), sigmoid scaled to $[0, \pi]$, linear $4 \to 4$ map (20 parameters), layer normalization, linear output.
LSTM+QNN	Same as the control with the linear map replaced by a 4-qubit circuit (two repetitions of RY/RZ encoding followed by a strongly entangling layer; Pauli-Z expectations; 24 trainable quantum parameters; `default.qubit`, backpropagation).
Parameter counts	`LSTM+QNN`: 2,368 (LSTM) + 208 (projection $51 \to 4$) + 8 (layer norm) + 5 (output) + 24 (circuit) = 2,613, of which 24 quantum (0.9 %); Linear-4D control: the same classical parts with the 20-parameter linear map instead of the circuit = 2,609.
Neural training	AdamW (learning rate 5×10^{-3}, weight decay 10^{-4}), learning-rate reduction on plateau (patience 3, factor 0.5), binary cross-entropy with positive-class weight, batch size 16, at most 25 epochs, gradient clipping at 1.0, double precision.
Selection	Early stopping on validation loss (patience 6, minimum improvement 10^{-4}); the checkpoint with the lowest validation loss is restored.

S1.4 Specification of the QML Core Experiment

Equal capacity or tuning effort across the four models should not be assumed.

S1.5 Simulation: All Scenarios

Table S4: All simulation rows. Population difference (Monte Carlo SE of the reference), rejection rate of $\Delta = 0$ (number of rejections in brackets; DeLong, block bootstrap), coverage of the population difference, mean of the estimates, SD of the estimates and mean standard errors (DeLong, block). Scenario B first run: the Monte Carlo standard error of the reference was not recorded. Scenario A values are those of the first script run, rounded.

Δ_{true}	Rej. DeLong	Rej. block	Cov. DeLong	Cov. block	Mean $\hat{\Delta}$	SD($\hat{\Delta}$)	SE DeLong	SE block
Scenario B: shared year effects ($\tau_\pi = \tau_s = 0.3$)								
0.0000 (0.0002)	0.040 (20)	0.074 (37)	0.960	0.926	0.0007	0.0108	0.0111	0.0108
0.0145 (0.0001)	0.318 (159)	0.350 (175)	0.946	0.934	0.0151	0.0115	0.0110	0.0107
0.0291 (0.0002)	0.738 (369)	0.778 (389)	0.946	0.936	0.0284	0.0109	0.0110	0.0107
Scenario C: + scorer-specific year effects ($\tau_m = 0.3$)								
0.0000 (0.0001)	0.068 (34)	0.076 (38)	0.932	0.924	-0.0008	0.0154	0.0144	0.0153
0.0136 (0.0001)	0.210 (105)	0.216 (108)	0.912	0.928	0.0161	0.0161	0.0144	0.0153
0.0276 (0.0002)	0.496 (248)	0.468 (234)	0.916	0.932	0.0272	0.0165	0.0142	0.0151
Scenario D: C with unequal year sizes								
0.0000 (0.0001)	0.096 (48)	0.084 (42)	0.904	0.916	0.0002	0.0168	0.0144	0.0155
0.0140 (0.0001)	0.202 (101)	0.168 (84)	0.900	0.910	0.0132	0.0171	0.0143	0.0154
0.0280 (0.0002)	0.506 (253)	0.480 (240)	0.882	0.920	0.0276	0.0172	0.0142	0.0154
Scenario E: all $\tau = 0.6$								
0.0000 (0.0002)	0.220 (110)	0.074 (37)	0.780	0.926	-0.0012	0.0290	0.0185	0.0273
0.0114 (0.0002)	0.262 (131)	0.122 (61)	0.784	0.936	0.0154	0.0290	0.0185	0.0277
0.0230 (0.0002)	0.350 (175)	0.206 (103)	0.752	0.908	0.0231	0.0305	0.0182	0.0272
Scenario B, first run (seed 0, 400 replications, $B = 300$)								
0.0004 (–)	0.037 (15)	0.080 (32)	0.958	0.925	0.0012	0.0109	0.0110	0.0108
0.0100 (–)	0.185 (74)	0.217 (87)	0.927	0.897	0.0104	0.0120	0.0110	0.0107
0.0190 (–)	0.430 (172)	0.470 (188)	0.970	0.935	0.0199	0.0114	0.0110	0.0106
0.0294 (–)	0.760 (304)	0.780 (312)	0.950	0.925	0.0290	0.0110	0.0110	0.0108
0.0391 (–)	0.950 (380)	0.953 (381)	0.945	0.925	0.0384	0.0114	0.0109	0.0106
Scenario A: $\rho = 0.75$, shared year effects (500 replications, $B = 500$; rates rounded to 2 decimals)								
0.0008 (–)	0.05	0.07	0.95	0.94	–	0.0165	0.0172	0.0166
0.0092 (–)	0.09	0.12	0.93	0.90	–	0.0179	0.0171	0.0167
0.0182 (–)	0.19	0.23	0.93	0.91	–	0.0179	0.0171	0.0165
0.0281 (–)	0.39	0.44	0.95	0.94	–	0.0166	0.0170	0.0165
0.0389 (–)	0.65	0.67	0.95	0.94	–	0.0169	0.0170	0.0164
0.0483 (–)	0.80	0.80	0.95	0.92	–	0.0171	0.0170	0.0166

S1.6 Hybrid Architectures and Further Comparisons

Table S5 gives the 15-seed AUC of all models next to the climatology. Random Forest has the highest mean in every pipeline and every hybrid is below it, a point-estimate ordering that is consistent across seeds. The ordering is a point-estimate result with limited evidential weight. The largest gap between Random Forest and any hybrid was 0.031 with the train-only Random Forest and 0.033 with the canonical one, in both cases against the same canonical-pipeline hybrids, which were not re-run train-only. For the hybrids, only seed-0 test probabilities were stored, so no paired comparison with Random Forest is made; the gaps above are point estimates. Fitting Random Forest on the 2,273 rows used by the neural models gives 0.6802 ± 0.0035 against 0.6851 ± 0.0026 on 2,787 rows: the difference of 0.005 is consistent with a training-size effect but lies well inside the noise.

Single-run claims among hybrids. In a single run, `LbL-QNN` reached an AUC of 0.672, tracking Random

Table S5: 15-seed test AUC, sorted by mean. Pipeline: TO = train-only anomaly standardization (re-run, primary); C = canonical pipeline. SD is the sample standard deviation ($n - 1$). Last column: difference to the climatology (AUC 0.6796, shown as 0.680 elsewhere); it is a descriptive gap, not a test.

Model	Pipeline	Mean AUC	SD	Gap to clim.
Random Forest (2,787 rows)	C	0.6851	0.0026	+0.006
Random Forest (2,273 rows)	TO	0.6846	0.0036	+0.005
Random Forest (2,787 rows)	TO	0.683	0.0035	+0.003
Random Forest (2,273 rows)	C	0.6802	0.0035	+0.001
Gradient Boosting	C	0.6692	0.0022	−0.010
Reuploading-3×4D	C	0.6688	0.0085	−0.011
Standalone LSTM	TO	0.668	0.0063	−0.011
Standalone LSTM	C	0.6657	0.0091	−0.014
Amplitude-16D	C	0.6643	0.0114	−0.015
XGBoost	TO	0.662	0.0054	−0.018
LbL-QNN	C	0.6605	0.0089	−0.019
LSTM+QNN	C	0.6589	0.0153	−0.021
XGBoost	C	0.6575	0.0051	−0.022
LSTM+PI-QNN	C	0.6555	0.0172	−0.024
LSTM+QTL	C	0.6519	0.0128	−0.028
Bottleneck-free LSTM (hidden 8)	C	0.6435	0.0065	−0.036

Forest (DeLong $p = 0.2124$) and nominally above `LSTM+QNN` ($p = 0.0070$) and the standalone LSTM (0.653, $p = 0.0263$; the latter comparison is also confounded by the capacity mismatch). Across 15 seeds the means of `LbL-QNN` and `LSTM+QNN` are 0.6605 and 0.6589. Over the $15 \times 15 = 225$ seed pairs, `LbL-QNN` has the higher AUC in 52.0 % of cases and `LSTM+QNN` in 48.0 %, an almost symmetric distribution. A paired DeLong test on every seed pair would need the per-seed test probabilities, which were archived only for seed 0, so none is made. An omnibus Friedman test finds no difference among the four quantum models ($\chi^2(3) = 2.600$, $p = 0.458$, Kendall's $W = 0.058$). These are seed-level statements; the sampling-error argument of Section 4.1 applies to them as well.

Equivalence margins. In the historical fixed-test seed-level analysis, GELU satisfied the ±0.010 margin and the linear controller satisfied ±0.015; LeakyReLU required ±0.020 (for `LSTM+QNN` versus `LbL-QNN`: mean difference −0.0016, 90 % CI $[-0.0095, +0.0062]$, $p_{\mathrm{TOST}} = 0.041$ at ±0.010). These tests use seed variance only, the margins were not justified by a scientific or practical criterion, and the paired structure of the seed index is a convention. They do not establish sampling-aware equivalence of predictive performance; a TOST that includes sampling error would require the per-seed test probabilities of the hybrids, which are not available (Sections S2.11 and S3.4).

Bottleneck-matched controls. The end-to-end hybrids compress a 51-dimensional vector to 4 angles. Three classical controllers on the identical 4-dimensional interface (linear, 20 parameters; LeakyReLU, 28; GELU, 180) obtain AUC 0.6545, 0.6516 and 0.6563 against 0.6589 for `LSTM+QNN`, with Wilcoxon $p \in [0.17, 0.64]$ (Section S2.11). The gaps are 0.003–0.007 on the historical seed-level data. A bottleneck-free LSTM with the same hidden size scores lower (0.6435 ± 0.0065) than the 4-dimensional controls, and full-feature XGBoost (0.662 train-only; 0.6575 in the canonical pipeline of the hybrids, Table S5) is comparable to `LSTM+QNN` (0.659). We therefore *withdraw* the explanation that the 4-dimensional bottleneck is the cause of the gap between hybrids and tree ensembles: the data do not identify the bottleneck as its cause, and removing the bottleneck did not raise the score of the small recurrent backbone.

Encoding bandwidth. Widening the interface to 16D amplitude embedding or 12D data re-uploading gives AUC 0.6643 and 0.6688, i.e. 0.005–0.010 above `LSTM+QNN`. The Friedman test over six quantum models is nominally significant ($p = 0.0228$, $W = 0.174$), but the advantage of re-uploading over `LSTM+QNN` does not survive Holm correction ($p_{\text{Holm}} = 0.083$), both variants add 600–800 classical projection weights, and the re-uploading circuit is also deeper. Even the best bandwidth variant (re-uploading, 0.6688) stays 0.016 below Random Forest (0.6851), so the data neither show that wider encodings close the gap to the tree ensembles nor that they fail to. The paired sampling uncertainty required for an architecture-level conclusion was not reconstructed for these historical runs; the result is descriptive and exploratory (Section S3.6).

PCA-4 control for the quantum kernel. Fitting a four-component PCA on the training partition only (65.7 % of the variance of the 35 features), a classical RBF-SVM reaches AUC 0.669 (Brier 0.2219) against 0.639 (0.2283) for the `QKSVM`. Random Forest and Gradient Boosting on the same four components score lower (0.631 ± 0.003 and 0.623 ± 0.0005). The 0.030 gap in favor of the classical kernel is descriptive: the paired standard error of this specific pair was not computed, so the result is consistent with, but does not establish, a disadvantage of the specific quantum feature map relative to classical kernels, and it shows that the low-dimensional compression is not the explanation for the modest QKSVM score.

S1.7 Internal Variable Mapping

Internal Variable Mapping (`dana_*` → `acp_*`). To maintain strict meteorological precision while preserving compatibility with legacy repository scripts, variable names have been standardized in the manuscript to reflect their proxy nature. The internal naming convention maps as follows:

- `dana_flag` ⟶ `acp_flag`: Binary indicator for Autumn Convective Precipitation conditions at the target month ($i + 12$), defined by autumn calendar month, negative sunshine anomaly, and positive gust anomaly.
- `dana_all` ⟶ `acp_windowed`: Boolean mask over the 3,315 windowed samples indicating target-month ACP conditions ($n = 156$; the $n = 159$ quoted for the full panel in Section S2.2 also counts months that are never prediction targets, because the first 12 months of each station are used only as inputs).
- `dana_te` ⟶ `acp_test`: Boolean mask over the 528 test-partition samples ($n = 25$ flagged months, of which 20 exceeded P_{60} and 5 did not).
- `Rc-DANA` / `rc_dana` ⟶ `Rc-ACP`: Tail recall evaluated strictly over the $n = 20$ flagged test months with confirmed extreme precipitation ($y = 1$).
- `ia_racha` ⟶ Wind-gust anomaly relative to station-month climatology.
- `choque_clausius` ⟶ Seasonal thermal contrast proxy ($T_{i-1} - T_{i-4}$).
- `potencial_convectivo` ⟶ Evaporation–variability interaction proxy ($\text{ET}_0^{\text{JJA}} \times \sigma_p$).
- Other code tokens (e.g. `pmed`, `veloc_ms`, `vuln_costera`, `dias_extremos`) keep their original Spanish names so that the manuscript matches the released code; Table S6 gives their English definitions.

S2 Extended Methodology and Diagnostics

S2.1 Station Test-Partition Representativeness and Coverage Asymmetry

Due to the strict per-station intersection of valid dates across all five AEMET source variables in the ETL pipeline (Section 2.1), station records exhibit heterogeneous temporal spans. San Javier's usable daily record is truncated at 2010 (at its shortest-covered source, atmospheric pressure), yielding only 36 samples in the test partition (6.8% of 528). Similarly, Valencia's record terminates in 2012 (50 test samples, 9.5%). In contrast, Alicante and Barcelona maintain continuous records extending through February 2026 (221 test samples each, 41.9% each).

While all four stations are proportionally represented in the training partition (each contributing between 17.6% and 30.0% of samples; Table 1), test-partition metrics post-2012 are effectively dominated by Alicante and Barcelona. We emphasize that no station data were fabricated or artificially extended to fill these gaps: monthly aggregation and downstream anomaly indices are computed only over genuinely observed date ranges per station. Re-acquiring or cross-validating post-2010 and post-2012 records for San Javier and Valencia would materially strengthen future replications of this benchmark and is, in our view, the clearest data-availability gap identified in this study.

S2.2 Daily Aggregation and P_{60} Target Validation

To verify that P_{60} exceedance represents structured, persistent precipitation events rather than diffuse wetness, we analyzed the daily composition of P_{60}-exceeding months. ACP-attributed months ($n = 159$ across the full panel, of which $n = 114$ exceeded P_{60}; $\sim 28\%$ did not exceed P_{60}) contain significantly more daily extremes (6.07 ± 2.26) than ordinary P_{60} months ($n = 1{,}231$, 4.94 ± 2.24; Mann-Whitney $U = 89627$, $p < 0.0001$), while the fraction contributed by the wettest single day shows no detected difference (0.490 ± 0.166 vs. 0.509 ± 0.192; $p = 0.530$). In this descriptive analysis, ACP-flagged months contain more extreme-precipitation days; event duration, consecutiveness and synoptic attribution were not established. The daily threshold defining *dias_extremos* (90th percentile, per-station) was estimated from the full historical record rather than the training partition alone; applying it to both compared groups does not by itself show the absence of bias, and it departs from the strict train-only threshold protocol used for P_{60}/P_{90}.

S2.3 Monthly Coverage Audit

Monthly precipitation is the sum of the available daily values of a station-month, and months with at least 10 valid days are retained. Of the 3,371 station-months, 3,363 are retained, and 327 of them are incomplete (fewer valid days than days in the month). By partition, 173 of 2,321 training months (7.5 %), 31 of 514 validation months (6.0 %) and 123 of 528 test months (23.3 %) are incomplete; by station, Barcelona has 217 of 743 months incomplete (29.2 %), San Javier 44 of 715 (6.2 %), Valencia 33 of 869 (3.8 %) and Alicante 33 of 1,036 (3.2 %). Among retained months, 9 have 10–14 valid days, 18 have 15–19, 52 have 20–24 and 56 have 25–27; the coverage ratio (valid days over days in the month) has a 5th percentile of 0.935 and a minimum of 0.32.

Table 6 recomputes the P_{60} threshold and the labels under stricter criteria. Under ≥ 15 valid days no label changes; under ≥ 20 three change (none in test), under ≥ 25 eight (two in test) and keeping complete

months only changes 17 (two in test), with the test prevalence between 0.407 and 0.417. The last row scales each monthly sum by days in month over valid days, which assumes that missing days rain like observed ones; it is a sensitivity scenario and neither a validated correction nor a bound, and it moves the threshold from 28.0 to 28.4 and the test prevalence to 0.430. At the project threshold, 18 incomplete months (12 in the test partition, 2.3 % of its rows) change label between the plain and the corrected sum; the remaining flips of the last row come from the shift of the recomputed threshold.

(The corresponding table is Table 6 of the main text.)

S2.4 Train-Only Anomaly Standardization and Leakage Validation

The synoptic anomaly features (Section S2.8) were originally computed by centering each station-month observation on the mean and standard deviation of its own (station, calendar-month) group across the *full* 1938–2026 record. To test whether this choice introduces validation/test-period information into the training partition, we implemented a strict train-only variant in which reference statistics are computed exclusively from observations dated on or before 1996, and applied identically across partitions:

$$z_{i,c,m} = \frac{x_{i,c,m} - \mu^{\text{train}}_{c,m}}{\sigma^{\text{train}}_{c,m} + \epsilon}, \qquad \epsilon = 10^{-4} \tag{S1}$$

where $\mu^{\text{train}}_{c,m}$ and $\sigma^{\text{train}}_{c,m}$ denote the mean and standard deviation of feature x for station c and calendar month m computed using only observations with year ≤ 1996. Station-month combinations absent from the training period default to $z = 0$. This train-only variant was substituted across all downstream synoptic anomaly features and their lagged derivatives.

To rigorously benchmark this effect on the canonical data build (`01_data.pt`, $P_{60} = 28.1$, $P_{90} = 86.32$ mm), we executed a 15-seed replication rerun of the primary baselines (Random Forest, XGBoost, and Standalone LSTM) under both the canonical full-record standardization and the strict train-only anomaly standardization. The rerun reproduced the canonical pipeline outputs before replacing the anomaly-standardization step; this served as a bit-for-bit verification of the evaluation script.

Evaluating paired differences (train-only minus canonical) across the 15 optimization seeds yields AUC deltas of:

- **Random Forest:** $\Delta\text{AUC} = -0.0022 \pm 0.0028$ (Wilcoxon signed-rank $p = 0.0353$);
- **XGBoost:** $\Delta\text{AUC} = +0.0041 \pm 0.0067$ (Wilcoxon $p = 0.0125$; Holm-Bonferroni corrected across three models $p_{\text{Holm}} = 0.037$);
- **Standalone LSTM:** $\Delta\text{AUC} = +0.0025 \pm 0.0090$ (Wilcoxon $p = 0.107$).

These paired p-values compare seeds on a single fixed test set and omit the test-set sampling error (standard error of one AUC about 0.024; Section 4.1); we therefore use them only descriptively. All AUC differences are small (about 0.004 AUC or less) and of mixed sign. Crucially, the nominally significant positive delta for XGBoost (+0.0041) goes in the direction *opposite* to leakage inflation: performance slightly improves under train-only standardization rather than degrading upon removing future information. Furthermore, an earlier exploratory check on a rebuilt dataset yielded an XGBoost delta of −0.0016 ($p = 0.17$), indicating that the sign and magnitude of this effect are not stable across dataset builds.

We therefore do not characterize this outcome as showing "no significant difference" in a strict formal sense; rather, the observed preprocessing differences were small and of mixed sign in this benchmark. These results do not establish that full-record preprocessing is free of leakage, and they do not isolate the differences as seed noise.

Compositional Shift in the Extreme Convective Evaluation Subset: While model discriminative ability remains stable, recomputing anomalies strictly from pre-1997 data noticeably shifts the threshold-based diagnostic subset *acp_flag* (identifying 176 panel events vs. 159 canonical). In the test partition, the train-only anomaly calculation flags 32 test months (25 with $y = 1$), compared to 25 flagged months (20 with $y = 1$) under the canonical full-record definition. Comparing the two test subsets reveals substantial churn: 23 months overlap, 2 canonical months leave the flagged set, and 9 newly flagged months enter. (The corresponding export field is `train_only_acp_test_n_positive`= 25 and counts only the flagged months with $y = 1$; the total flagged count is 32). Evaluating Rc-ACP naively by pairing the model outputs across differing masks ($n = 25$ vs. $n = 20$) is methodologically invalid. When evaluated strictly on the canonical test mask ($n = 20$), the mean extreme-recall deltas (train-only minus canonical) are: Random Forest ΔRc-ACP $= -0.010$, XGBoost ΔRc-ACP $= -0.070$, and Standalone LSTM ΔRc-ACP $= -0.013$.

Declaration of Training Set Sizes and Fitting Protocol: Code audit of the replication script and artifact checkpoints confirms that both the canonical and train-only scikit-learn models (Random Forest and XGBoost) were fitted on exactly 2,787 rows (the train + validation partition $X_{\text{sk}} = X_{\text{all}}[: s_2]$, spanning 1938–2007; see Section 2.2), while Standalone LSTM was fitted on exactly 2,273 rows (the training partition) with early stopping on 514 rows (the validation partition). The paired anomaly comparison ($\Delta\text{AUC} \leq 0.004$) is therefore strictly controlled for fitting-partition size across both pipeline conditions (2,787 rows for tree ensembles; 2,273 rows for LSTM), eliminating fitting-size asymmetry as a confounding factor in the leakage assessment. Separately, when canonical Random Forest is fitted strictly on the 2,273 training-only rows (matching the neural fitting partition without validation data), its 15-seed performance is 0.6802 ± 0.0035 (range $[0.6732, 0.6867]$), which is consistent with a small training-size contribution (~ 0.005 AUC) to the classical ensemble margin but lies well inside the noise of this benchmark.

S2.5 Activation-Specific Collapse in Narrow Classical Bottlenecks

During development of the narrow MLP control, 9 of 15 seeds collapsed to constant prediction (AUC = 0.500). This was diagnosed as a dead-ReLU pathology, where all hidden nodes fall into negative pre-activation regimes during random initialization from $\mathcal{N}(0, \sigma^2)$. Replacing ReLU with LeakyReLU (slope = 0.01) and He initialization, or using a bare linear layer, completely resolved the collapse across all 15 seeds, indicating that the pathology was associated with the activation rather than with capacity. The quantum module does not contain ReLU units; this does not imply immunity of the hybrid model to degenerate predictions or optimization failure.

S2.6 Layer-by-Layer Depth Justification

The `LbL-QNN` model uses a 3-layer circuit to facilitate a non-degenerate freezing schedule across three training stages (Skolik et al., 2021). To rigorously isolate the effect of the LbL protocol from the confounding effect of circuit depth, we additionally trained a depth-matched control, `LSTM+PI-QNN-3L`: an end-to-end (non-frozen) 3-layer circuit, replicated across the same 15 seeds.

Depth alone (2-layer → 3-layer, both end-to-end) produced no statistically significant change in either variance (Levene's test, $F_{\text{Levene}} = 0.16$, $p = 0.693$) or mean AUC (Wilcoxon, $p = 0.208$) in this historical comparison. The depth-matched comparison showed lower observed across-seed variability for the combined layer-wise training configuration ($\sigma = 0.0089$ vs. $\sigma = 0.0154$; Levene's test, $F_{\text{Levene}} = 7.47$, $p = 0.011$; mean AUC Wilcoxon $p = 0.073$). The design does not isolate layer-wise scheduling from the penalty, training duration and learning-rate changes: an unpenalized layer-by-layer control and an epoch-matched (55-epoch) end-to-end control were not evaluated.

For full transparency, the originally reported comparison between `LbL-QNN` and the 2-layer `LSTM+PI-QNN` baseline ($\sigma = 0.0089$ vs. $\sigma = 0.0171$)—which conflates both depth and protocol—does not reach significance under a formal Levene's test ($F_{\text{Levene}} = 1.80$, $p = 0.191$) despite the descriptive point-estimate gap. The variance-stabilizing effect of `LbL-QNN` is therefore supported by the depth-matched comparison above, not by the uncontrolled 2-layer-vs-3-layer comparison.

S2.7 Encoding-Bandwidth Architectures

Both extension architectures use separate projection layers mapping the 51-dimensional pre-bottleneck representation $\tilde{\mathbf{h}}_t \in \mathbb{R}^{51}$ (16D LSTM hidden state concatenated with the 35D raw engineered feature vector) directly to higher-dimensional spaces, independently of and in parallel to the baseline architectures' own 4D projection head:

1. **Amplitude Embedding:** Maps $\tilde{\mathbf{h}}_t$ to $\mathbf{z} \in \mathbb{R}^{16}$ via a dedicated linear layer. The state is prepared as $|\psi\rangle = \sum_{i=0}^{15} z_i |i\rangle$, with $\sum |z_i|^2 = 1$.
2. **Data Re-uploading:** Maps $\tilde{\mathbf{h}}_t$ to $\theta \in \mathbb{R}^{12}$ via a dedicated linear layer, partitioned into three disjoint 4-dimensional subsets injected sequentially as rotation angles across the three VQC layers.

S2.8 Engineered Meteorological Features and Glossary

To capture antecedent physical conditions without future-to-past leakage, we construct exactly 35 tabular covariates per station-month from lagged observations: 12 antecedent precipitation lags ($P_i, \ldots, P_{i+11}$; the target month is $i + 12$), 17 physical and synoptic indicators (including 3 monthly anomaly indicators centered against station-month climatology, evapotranspiration indicators, and instability proxies), 2 cyclic seasonal encodings (sin, cos of target month $i + 12$), and 4 one-hot station identifiers (`ciudad_*`). Table S6 provides a complete enumeration of all 35 features, their mathematical definitions, and repository code tokens.

These covariates provide candidate inductive biases and input features, not physical state variables in a numerical weather prediction model. The empirical correlation plots, post-hoc latent-coordinate mapping, and detailed formulation of the heuristic thermodynamic loss function are presented in Section S7.

S2.9 Experimental Design and Architecture Figures

The modular pipeline connecting the recurrent feature extractor, the 4-dimensional projection bottleneck, and the variational quantum circuit is schematized in Figure S2, and the multi-seed evaluation hierarchy across all classical and hybrid tiers is outlined in Figure S3.

Table S6: Complete enumeration of the 35 engineered features (12 lags + 17 physical + 2 cyclic + 4 station indicators).

Group	Feature Label	Code Token	Definition / Formulation
Antecedent	Precip. Lags (12)	`precip_lag_1..12`	Monthly precipitation sums for months $i, \ldots, i-11$
	Pressure variability	`dp_std`	Standard deviation of daily sea-level pressure in month i
Physical &	Mean pressure	`pmed`	Monthly-mean sea-level pressure (hPa)
	Diurnal temp. range	`t_rango`	Mean daily maximum minus minimum temperature (°C)
	Mean wind speed	`veloc_ms`	Monthly-mean surface wind speed (m/s)
	Evapotranspiration	`et0_sum`	Accumulated reference evapotranspiration (FAO-56)
	Summer evapotransp.	`et0_verano_prev`	Antecedent summer (JJA) accumulated ET_0
	Evapotransp. Lags	`et0_lag1m, et0_lag3m`	1-month and 3-month lagged ET_0 sums
	Evaporative deficit	`fao_evapo_deficit`	Antecedent ET_0 minus antecedent precipitation sum
	Coastal vulnerability	`vuln_costera`	Coastal proximity indicator interacting with barometric deficit
Anomalies	Pressure anomaly	`iap`	Monthly-mean pressure minus station-month climatology
	Wind-gust anomaly	`ia_racha`	Maximum gust speed minus station-month climatology
	Temp. anomaly	`t_anomaly`	Monthly-mean temperature minus station-month climatology
Proxies	Instability index	`ib`	Barometric instability index: σ_p/t_rango
	Insolation ratio	`iir`	Sunshine duration / theoretical extraterrestrial maximum
	Thermal contrast proxy	`choque_clausius`	3-month lagged temperature anomaly difference ($T_{i-1} - T_{i-4}$)
	Convective potential	`potencial_convectivo`	Summer $ET_0^{JJA} \times$ monthly pressure variability (σ_p)
Seasonal	Target month cyclic	`mes_sin, mes_cos`	$\sin(2\pi m/12)$ and $\cos(2\pi m/12)$ for target month $i+12$
Station	Station Indicators	`ciudad_*`	One-hot indicators (Alicante, Barcelona, San Javier, Valencia)

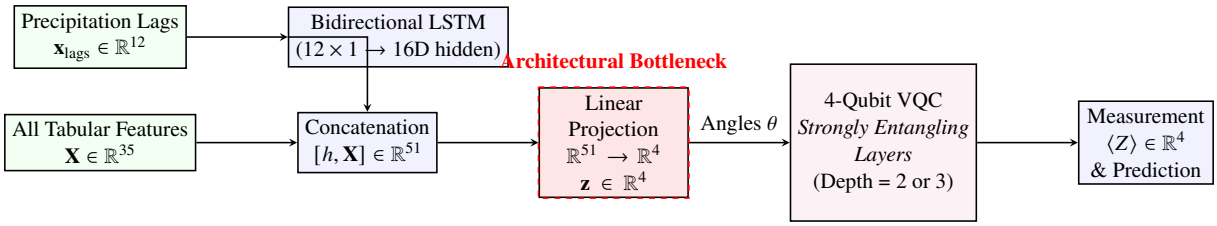


Figure S2: Hybrid Quantum-Classical Architecture. The bidirectional LSTM processes the univariate 12-month precipitation lag sequence ($\mathbf{x}_{\text{lags}} \in \mathbb{R}^{12}$), producing a 16-dimensional recurrent representation. This is concatenated with the full 35-dimensional tabular feature vector (which redundantly contains the 12 lags alongside physical covariates and station indicators) to form a 51-dimensional representation. A classical linear projection ($\mathbb{R}^{51} \rightarrow \mathbb{R}^4$) then compresses the features into 4 continuous rotation angles for encoding into the 4-qubit Variational Quantum Circuit.

To test whether domain-motivated soft constraints alter optimization trajectories, we evaluate two hybrid configurations incorporating heuristic thermodynamic regularization (`LSTM+PI-QNN` and `LbL-QNN`). As detailed in Section S7, this scheme supplements binary cross-entropy with two phenomenological soft penalties: a temperature-dependent gating term ($\mathcal{L}_{\text{CC}}$) penalizing high precipitation predictions under restricted saturation-capacity conditions, and a latent-coordinate balance term ($\mathcal{L}_{\text{hMSE}}$) acting on the 4-qubit expectation readout coordinates ($q_0, \ldots, q_3$). These are empirical soft regularizers rather than first-principles thermodynamic conservation laws. As shown in Section S3.4, 15-seed replication reveals that this heuristic regularization confers no statistically significant performance advantage over the unconstrained hybrid baseline (Friedman $p = 0.458$).

S2.10 Quantum and Classical Architectures

We evaluate ten distinct model architectures alongside one threshold-matched operational configuration (yielding eleven evaluation configurations in Table S8), grouped into four paradigms:

1. **Statistical and Climatological Reference Baselines:** We evaluate non-parametric reference benchmarks alongside L_2-regularized Logistic Regression: (i) *Month* × *Station Climatology*, which assigns each test sample the empirical frequency of precipitation exceeding P_{60} observed for that specific station and calendar month in the training partition (AUC = 0.680, Brier score = 0.222, Brier skill score relative to the sample base rate = +0.082); and (ii) a binarized month×station climatology, evaluated as an unranked binary baseline thresholded at 0.5 (Acc = 64.58%, F1 = 0.522, binary AUC = 0.619), whereas physical lag-1 precipitation persistence achieves AUC = 0.589.

2. **Classical Ensembles:** Random Forest (300 estimators), Gradient Boosting (200 estimators), and XGBoost (500 estimators, depth=4).

3. **Deep Learning Baselines:** A standalone bidirectional LSTM (`PureLSTM`) with 2 layers and 16 hidden units per direction (yielding a concatenated 32-dimensional sequence representation), concatenated with the 35 input features (67 dimensions), followed by batch normalization (`BatchNorm1d(67)`) and a two-layer classification head (`Linear(67, 16)` + LayerNorm + `Linear(16, 1)`), totaling 10,103 parameters, optimized using AdamW. Alongside this model, we evaluate `Matched LSTM`, an operational configuration of `PureLSTM` with its classification threshold adjusted to match the false positive rate of the hybrid models ($\tau = 0.531$; Section S4). We emphasize that the standalone LSTM is not capacity-matched to the hybrids (10,103 vs. ~2,600 parameters; 16

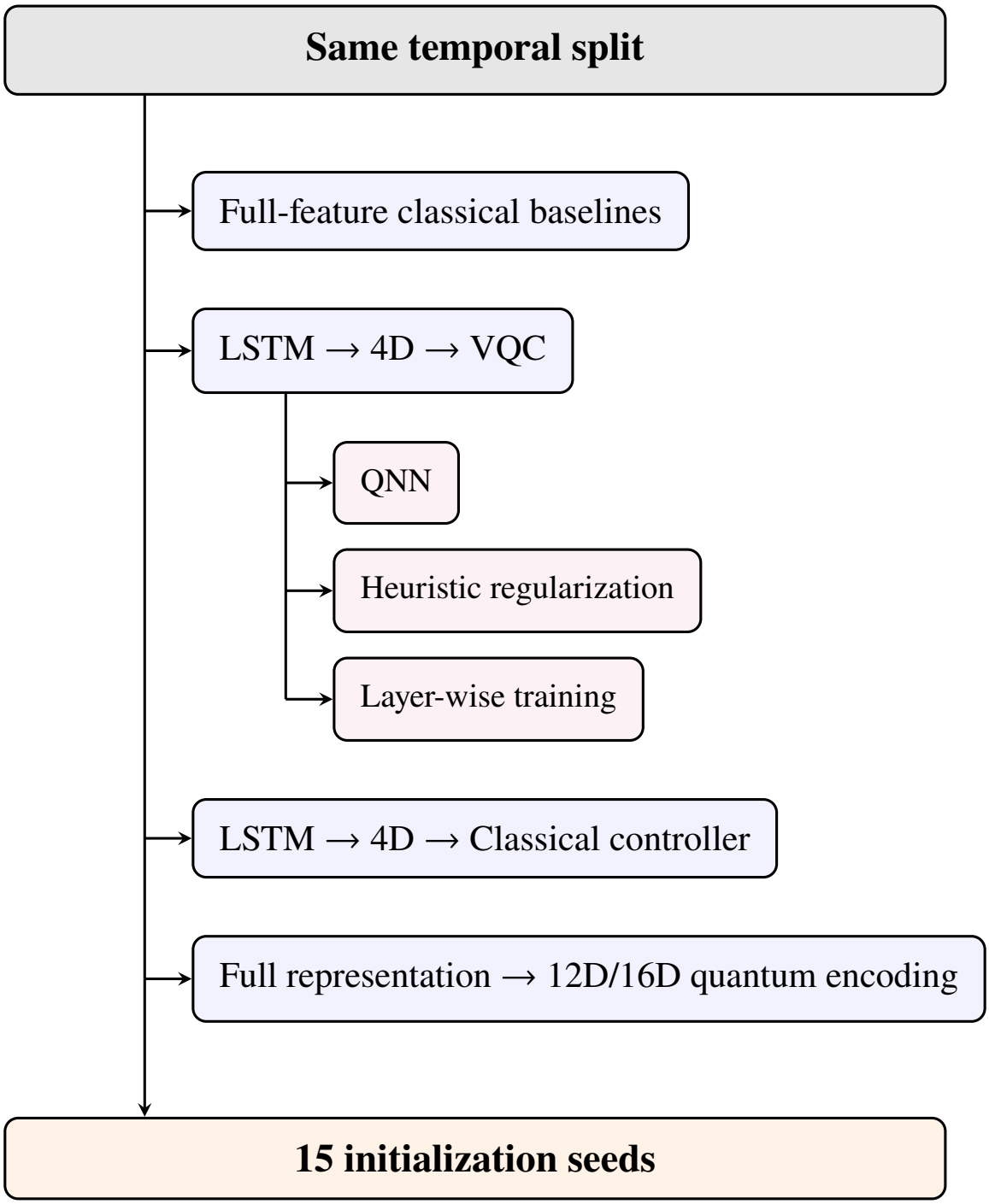


Figure S3: Schematic of the experimental design. All evaluated architectures originate from the identical temporal split and culminate in a rigorous 15-seed parameter initialization protocol to control for optimization variance. The hierarchy explicitly isolates the full-feature classical baselines from the hybrid quantum models, and pairs the latter with bottleneck-matched classical controls and encoding-bandwidth ablations.

vs. 8 hidden units per direction; batch normalization; and direct access to raw features without a 4-dimensional projection bottleneck).

4. **Hybrid Quantum-Classical Models:** Developed using PennyLane's `default.qubit` simulator. `LSTM+QNN`, `LSTM+PI-QNN` and `LbL-QNN` use a bidirectional LSTM trained end to end (hidden dimension 8 per direction, 2,368 parameters) concatenated with the 35 input features to produce a 51-dimensional representation, which is compressed via a linear layer (`Linear(51, 4)`, 208 parameters) into a 4-dimensional projection bottleneck before being encoded into a 4-qubit Variational Quantum Circuit. The circuit uses a Strongly Entangling Layers ansatz with same-vector data re-uploading (R_y and R_z rotations per qubit repeated in each layer; Pérez-Salinas et al., 2020) followed by parameterized SU(2) Rot gates and a circular CNOT entanglement topology ($L = 2$ layers with 24 quantum parameters for baseline models; $L = 3$ layers with 36 quantum parameters for the layer-wise `LbL-QNN` model; see Figure S4). We distinguish this repeated encoding from the *information-widening* re-uploading variant introduced later (Section S3.6), which injects distinct feature subsets at each block.

    - `LSTM+QNN`: Standard end-to-end unconstrained variational network (2 layers).
    - `LSTM+PI-QNN`: End-to-end network optimized with the heuristic thermodynamic loss $\mathcal{L}_{\text{total}}$ (2 layers; Section S7).
    - `LSTM+QTL`: Quantum Transfer Learning (Mari et al., 2020), which differs in architecture from the three models above: the standalone `PureLSTM` (10,103 parameters, Table S18) is first trained as a classical model with early stopping and then frozen; its 32-dimensional state and the 35 features (67 dimensions) feed a trainable classical adapter ($67 \to 8 \to 4$), followed by a single angle encoding (RY, no re-uploading) and a 2-layer circuit. It has 617 trainable parameters, of which 24 are quantum.
    - `LbL-QNN`[1]: applies the layer-by-layer training protocol to the `LSTM+PI-QNN` architecture, using a 3-layer circuit to support a non-degenerate three-stage freezing schedule.
    - `QKSVM`: Quantum Kernel Support Vector Machine utilizing a Nyström approximation ($M = 64$ landmarks) to handle the panel's size.

S2.11 Bottleneck-Controlled Classical Baseline

A central confound in comparing hybrid quantum architectures against full-feature classical ensembles is that the VQC-based models operate on a 4-dimensional projection produced by the LSTM bottleneck, whereas classical ensembles ingest the full 35-dimensional engineered feature space, including geographic fixed effects.

[1]This architecture combines layer-by-layer training (Skolik et al., 2021) with the heuristic thermodynamic loss (Section S7). As shown in Section S2.6, a depth-matched ablation specifically evaluating `LbL-QNN` against an end-to-end 3-layer architecture (`LSTM+PI-QNN-3L`) showed lower observed across-seed variability for the combined layer-wise configuration. However, this comparison is confounded by an expanded training budget (55 total epochs: 10 epochs per layer across 3 layers = 30 epochs, followed by 25 global fine-tuning epochs vs. 25 single-run epochs) and a staged learning rate schedule (5×10^{-3} then 5×10^{-4} vs. fixed 5×10^{-3}). The observed variance stabilization is therefore associated with the combined multi-stage schedule and expanded budget rather than demonstrably caused by the layer-wise inductive bias in isolation, particularly since a penalty-free layer-by-layer control and an epoch-matched (55-epoch) end-to-end control were not evaluated; the observed performance should be attributed to the combined layer-wise/heuristic configuration.

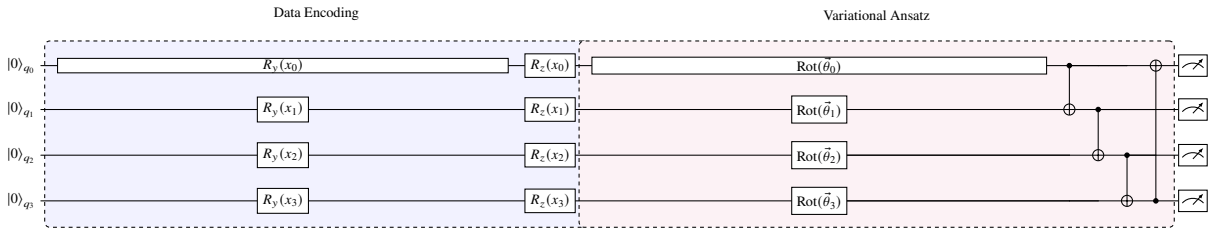


Figure S4: Gate-level schematic of a single layer of the 4-qubit Variational Quantum Circuit (VQC). The 4-dimensional classical bottleneck vector $\mathbf{x}$ is injected into the quantum state via consecutive $R_y(x_i)$ and $R_z(x_i)$ rotations on each qubit (Data Encoding; Pérez-Salinas et al., 2020). The trainable ansatz consists of general SU(2) rotations (Rot gates, each parameterized by a 3-component vector $\vec{\theta}_i = (\theta_{i,1}, \theta_{i,2}, \theta_{i,3})$) followed by a circular CNOT entanglement topology. To implement the Strongly Entangling Layers architecture, this entire encoding-ansatz block is repeated L times (with $L = 2$ for the baseline models, yielding 24 quantum parameters, and $L = 3$ for the layer-wise `LbL-QNN` model, yielding 36 quantum parameters). Readout is performed via Pauli-Z measurements on all qubits.

To isolate this confound, we evaluate three bottleneck-matched classical control configurations (`LSTM+MLP-4D`) spanning a range of parameter counts. These architectures are identical to `LSTM+QNN` in their upstream feature extractor, 4-dimensional projection bottleneck, and downstream classification head, but replace the 4-qubit VQC with a classical controller (linear, LeakyReLU, or GELU MLP). All configurations were replicated across the same 15 seed-indexed initializations used for `LSTM+QNN` (Table S7). An initial width-2 ReLU-based control collapsed to chance-level output (AUC= 0.500) in 9 of 15 seeds due to dead-unit failure; we resolved this with the linear and LeakyReLU configurations reported below, which is consistent with an activation-specific rather than capacity-related failure (full diagnostic in Section S2.5).

We emphasize that parameter count is a coarse and imperfect proxy for functional capacity: a linear model and a nonlinear, GELU-activated MLP with comparable parameter counts are not necessarily comparable in expressive power, since capacity depends jointly on activation nonlinearity and architecture, not on parameter count alone. We therefore describe these three controls as spanning a range of parameter counts and architectural nonlinearities, rather than as strictly bottleneck-controlled to the quantum circuit in a formal sense. The qualitative conclusion below is nonetheless robust to this caveat, as the absence of a statistically significant difference from the quantum circuit holds independently for both the least expressive (linear) and most expressive (GELU) control considered.

A 4-qubit circuit with 24 variational parameters operates on a state space of dimension $2^4 = 16$; its statevector evolution is represented by a 16×16 unitary matrix, which is classically simulable by statevector methods. Parameterized quantum circuits with angle encoding compute truncated Fourier series of the input coordinates (Schuld & Killoran, 2022); classical models with trigonometric activations could therefore be functional analogues, but this was not tested here.

As a historical fixed-test seed-level diagnostic (canonical pipeline), we conducted Two One-Sided Tests (TOST) against `LSTM+QNN` across the 15 seeds. Under a reference margin of $\Delta = \pm 0.010$ AUC, `GELU MLP-4D` satisfies the margin (Diff $= -0.0025$, paired 90% CI $[-0.0092, +0.0041]$, $p_{\mathrm{TOST}} = 0.0350$). The smaller `Linear-4D` (Diff $= -0.0044$, 90% CI $[-0.0121, +0.0032]$, $p_{\mathrm{TOST}} = 0.1113$) and `LeakyReLU-4D` (Diff $= -0.0073$, 90% CI $[-0.0151, +0.0005]$, $p_{\mathrm{TOST}} = 0.2772$) do not reject the ± 0.010 margin. However, under a sensitivity margin of $\Delta = \pm 0.015$ AUC, `Linear-4D` also satisfies the margin ($p_{\mathrm{TOST}} = 0.0148$),

Table S7: Bottleneck-controlled comparison: classical controls versus the quantum circuit under an identical 4-dimensional information bottleneck, versus the full-feature Random Forest reference, sorted descending by mean AUC. All values are 15-seed distributions with seed-indexed initialization (the index is a label, not a pairing device).

Architecture	Controller params	Mean AUC	Std. Dev.	Max
Random Forest (train+val, 2,787 rows)*	Ensemble	0.6851	0.0026	0.690
Random Forest (matched, 2,273 rows)*	Ensemble	0.6802	0.0035	0.687
LSTM+QNN (quantum, 4-dim. bottleneck)	24	0.6589	0.0153	0.681
GELU MLP (classical control, 4-dim. input)	180	0.6563	0.0071	0.6725
Linear (classical control, 4-dim. input)	20	0.6545	0.0069	0.6680
LeakyReLU MLP (classical control, 4-dim. input)	28	0.6516	0.0083	0.6636

*Tree ensemble capacity is determined by the number of leaves/nodes rather than differentiable parameters; the 300-tree Random Forest with `max_depth=8` and the hybrid belong to different function classes and are not capacity-matched. The 2,273-row model provides the strictly matched training comparison to the neural and hybrid models. Parameter counts reported under Controller params reflect only the 4-dimensional controller/neck replacing the quantum circuit; for total model parameter accounting (including the shared 2,589 classical backbone), see Table S18.

and `LeakyReLU-4D` requires $\Delta = \pm 0.020$ AUC ($p_{\text{TOST}} \leq 0.0064$ for all three controllers). Wilcoxon signed-rank tests similarly show no significant differentiation ($p \in [0.17, 0.64]$), and do not detect a circuit effect at the fixed 4-dimensional interface. As in Section 4.5, these seed-level tests omit test-set sampling error; they do not establish sampling-aware equivalence of predictive performance, and all three paired 90% intervals lie within ±0.016 AUC.

We additionally evaluated classical controls under an unsupervised 4-dimensional compression of the entire 35-dimensional input space, providing an information-matched benchmark for `QKSVM` (Section 2.2). Fitting Principal Component Analysis strictly on the training partition (PCA(4), capturing 65.70% of total variance across the 35 features) and evaluating on the test partition, a classical Support Vector Machine with an RBF kernel achieves an AUC of 0.6686 and a Brier score of 0.2219 (with Acc = 62.31%, F1 = 0.5158, and Rc-ACP = 0.8500). This classical 4-dimensional benchmark outperforms the 4-qubit Quantum Kernel SVM (AUC = 0.6390, Brier score = 0.2283; Table S8) by 0.030 AUC (a descriptive gap; its paired sampling uncertainty was not computed), whereas tree ensembles on PCA-4 achieve lower discriminative performance (Random Forest: 0.6312 ± 0.0030; Gradient Boosting: 0.6230 ± 0.0005). This indicates that the modest score of `QKSVM` is not explained by low-dimensional compression alone; whether it reflects the inductive bias of the evaluated quantum feature map is not established by these data.

Random Forest is not a bottleneck-matched comparator for the VQC; it operates on the uncompressed 35-dimensional space with 300 trees, establishing an empirical tabular reference (Mean AUC = 0.6802±0.0035 on matched 2,273 rows; 0.6851 ± 0.0026 on 2,787 rows). The 4D controls hold the input interface constant, isolating circuit-specific effects from representation compression: at this fixed interface, the historical seed-level means of the controllers lie within about 0.007 AUC of the hybrid. While restricting models to a 4D interface degrades performance relative to Random Forest, the information bottleneck alone cannot explain the classical-over-quantum gap, because full-feature tree ensembles such as XGBoost (0.6575±0.0051) achieve discriminative performance comparable to baseline `LSTM+QNN` (0.6589±0.0153). The residual gap to Random Forest may reflect model-family capabilities on tabular data rather than an encoding bottleneck, and was not tested here.

As exploratory evidence on whether the 4-dimensional compression itself constrains the recurrent backbone, we also evaluated a bottleneck-free classical LSTM with the same hidden dimension 8 (`bottleneck_free_lstm`, arithmetically ~2,522 parameters: 2,368 LSTM + 102 BatchNorm + 52 linear head, though parameter counts were not stored in the checkpoint). Unlike the hybrid, which projects the 51-dimensional representation (16D LSTM hidden + 35D raw features) down to 4 dimensions before classification, this model attaches a direct linear classification head to the 51-dimensional vector. Across 15 seeds, it achieves an AUC of 0.6435 ± 0.0065. Because the classifier head differs and no paired seed-by-seed test was logged, this exploratory result does not support a claim that removing the bottleneck formally confirms compression as a performance limit. Descriptively, the bottleneck-free model falls below both the 4D linear control (0.6545 ± 0.0069) and the 4D MLP controls (0.6563 ± 0.0071, 0.6516 ± 0.0083), as well as the larger Standalone LSTM (0.6657 ± 0.0091). That the 4D linear control scores 0.011 above the unprojected 51D head (a descriptive gap) at most suggests that low-dimensional linear projection can act as a regularizer for this small recurrent backbone, rather than compression acting solely as an upper bound on performance. We explicitly separate the claim of projection regularization from the claim that quantum circuits provide an inductive advantage at that interface.

The training protocol reveals an important asymmetry between model families: tree ensembles (Random Forest, Gradient Boosting, and XGBoost) were originally fit on the 2,787 rows of the combined training and validation partitions ($X_{\text{sk}} = X_{\text{all}}[: s_2]$), whereas neural and hybrid quantum models were fit on the 2,273 training rows with validation rows reserved for early stopping (Section 2.2). Evaluating Random Forest strictly on the 2,273 training rows with canonical hyperparameters yields a 15-seed mean AUC of 0.6802 ± 0.0035 (range $[0.6732, 0.6867]$), compared to 0.6851 ± 0.0026 on 2,787 rows. An automated Optuna search (50 trials) on the 2,273 training rows yielded 0.6798 ± 0.0045. We report both: the primary Random Forest of the main text is the train-only pipeline fitted on 2,787 rows, and the 2,273-row model is a sensitivity analysis whose 0.005 lower AUC is within the noise of the benchmark.

S3 Additional Results

S3.1 Single-Run Performance

We evaluate model performance on the canonical spatiotemporal test partition ($N = 528$ station-months, 2008–2026) using the canonical dataset `01_data.pt` ($P_{60} = 28.1$, $P_{90} = 86.32$ mm). As established by our 15-seed replication rerun on the canonical dataset (Section S2.4), adopting a strict train-only standardization protocol yields downstream performance deltas of small magnitude (≤ 0.004 AUC) and mixed sign across model families, reflecting optimization seed variability rather than systematic leakage inflation.

Table S8 summarizes the single-run test performance across all evaluated models at a fixed initialization seed (`seed=42`). Global accuracy is misleading due to class imbalance; we therefore focus on F1-score, ROC-AUC, and Precision-Recall AUC (PR-AUC; Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015). The tree-based **Random Forest** establishes the classical reference across ROC-AUC and F1 (AUC = 0.688, F1 = 0.603), while the non-parametric **Month × Station Climatology** establishes an empirical baseline that matches or exceeds all quantum models (AUC = 0.680, PR-AUC = 0.603). The hybrid quantum-classical architectures cluster tightly below (AUC: 0.639–0.672; F1: 0.580–0.604).

Table S8: Comparative performance on the spatiotemporal test set (Single-Run), including Precision-Recall AUC (PR-AUC) alongside ROC-AUC. Single representative run evaluated at default threshold 0.5 ($n = 20$ ACP-flagged test months exceeding P_{60} out of 25 flagged total; $n = 49$ P_{90}-exceeding test months). Month × Station Climatology establishes the non-parametric empirical baseline.

Model	**Acc (%)**	**F1**	**ROC-AUC**	**PR-AUC**	**Rc-P_{90}**	**Rc-ACP**	**FPR**
Climatological Reference							
Month × Station Climatology	61.2	0.547	0.680	**0.603**	0.653	0.600	0.360
Statistical & Ensemble							
Logistic Regression	62.3	0.472	0.655	0.546	0.571	0.650	0.228
Random Forest	63.8	0.603	**0.688**	0.591	0.755	0.900	0.383
Gradient Boosting	61.6	0.522	0.671	0.577	0.592	0.700	0.312
XGBoost	59.1	0.548	0.650	0.545	0.694	0.850	0.418
Deep Learning							
Standalone LSTM	54.5	0.585	0.653	0.587*	0.816	0.950	0.617
Matched LSTM†	57.2	0.565	0.653	0.587*	0.776	0.950	0.502
Quantum-Classical Hybrid							
LSTM+QNN	58.0	0.599	0.639	0.504	0.796	0.950	0.550
LSTM+PI-QNN	58.9	0.583	0.662	0.558	0.755	0.950	0.489
LSTM+QTL	58.7	0.604	0.651	0.518	0.837	0.850	0.537
LbL-QNN	59.1	0.589	0.672	0.581	0.796	0.900	0.495
QKSVM	59.8	0.475	0.639	0.510	0.612	0.600	0.293

*PR-AUC is threshold-independent and therefore identical to the unmatched Standalone LSTM row; only the operational point (FPR, recall) changes under matching.

†Threshold-recalibrated variant of Standalone LSTM, evaluated at an operating point matched in false-detection rate to `LbL-QNN` ($\tau = 0.531$); defined and motivated in Section S4.

Rc-ACP is evaluated strictly over the $n = 20$ positive test months ($y = 1$) matching the Autumn Convective Proxy in the canonical pipeline (out of 25 flagged months total), distinct from the 32 flagged months (25 positive) identified under train-only anomaly standardization (Section S2.4); if all 25 flagged months are counted regardless of precipitation (the alarm rate), the respective values are: Climatology 0.640, Logistic Regression 0.600, Random Forest 0.880, Gradient Boosting 0.680, XGBoost 0.840, Standalone LSTM 0.960, Matched LSTM 0.960, LSTM+QNN 0.920, LSTM+PI-QNN 0.960, LSTM+QTL 0.840, LbL-QNN 0.920, QKSVM 0.520.

In this single realization, the layer-wise hybrid `LbL-QNN` achieves the highest quantum AUC (0.672), tracking Random Forest (Table S8) with no statistically significant difference under an uncorrected DeLong test (DeLong et al., 1988) ($p = 0.2124$; formal equivalence not tested). It nominally outperforms unconstrained `LSTM+QNN` ($p = 0.0070$) and `Standalone LSTM` ($p = 0.0263$), though the latter comparison is heavily confounded by the capacity mismatch (10,103 vs. ~2,600 parameters; Section 2.2).

Crucially, this apparent single-run superiority dissolves under closer examination. First, comparing Random Forest to Month × Station Climatology yields no statistically significant advantage under a paired DeLong test (ΔAUC = +0.0079, SE = 0.0177, $z = 0.449$, $p = 0.6534$). Accounting for temporal dependence via a year-block bootstrap ($B = 2{,}000$ resamples of the 19 test calendar years 2008–2026; Künsch, 1989), the difference remains non-significant (ΔAUC = +0.0068, 95% block CI: $[-0.0171, +0.0284]$, $p = 0.5420$). Thus, neither Random Forest nor any evaluated quantum architecture demonstrates statistically verifiable predictive skill beyond empirical seasonal-geographic climatology when temporal autocorrelation is respected.

Second, disaggregating test performance across individual monitoring stations reveals that the pooled ROC-AUC ($N = 528$) aggregates disparate regional regimes. In Alicante ($N = 221$, 29.4% positive rate), Random Forest achieves an AUC of 0.698 and `LbL-QNN` achieves 0.683, compared to 0.681 for Climatology. In Barcelona ($N = 221$, 50.2% positive rate), Climatology achieves an AUC of 0.663, outperforming both Random Forest (0.634) and all quantum models (`LbL-QNN`: 0.643; `LSTM+QNN`: 0.538). In Valencia ($N = 50$, 56.0% positive rate), Climatology (0.692) likewise exceeds Random Forest (0.690), `LbL-QNN` (0.649), and `LSTM+QNN` (0.607), while in San Javier ($N = 36$, 36.1% positive rate), all models cluster between 0.535 and 0.575. This indicates that a substantial fraction of pooled discriminative performance arises from separating high-base-rate from low-base-rate stations rather than dynamically resolving extreme months within a station regime.

Third, because the extreme tail contains limited test observations ($n = 20$ ACP months exceeding P_{60}, $n = 49$ P_{90} months), point estimates of extreme recall carry wide 95% Wilson confidence intervals (which ignore temporal and spatial dependence, and whose overlap is not a paired test of equality) that broadly overlap across all architectures (e.g., Random Forest 0.900 [0.699, 0.972] vs. `LSTM+QNN` 0.950 [0.764, 0.991]; documented in Section S4). Fourth, the apparent operational difference between `Standalone LSTM` (Rc-ACP = 0.950, FPR = 0.617) and `LbL-QNN` (Rc-ACP = 0.900, FPR = 0.495) is entirely an artifact of default threshold selection: recalibrating the classical LSTM to operate at an identical FPR of ≈50.2% (`Matched LSTM`, $\tau = 0.531$) preserves its recall at 0.950 (19/20) with no statistically significant difference under paired bootstrap resampling ($n = 2{,}000$; full analysis in Section S4). Finally, as demonstrated in Section S3.4, multi-seed replication reveals that these single-run margins reflect initialization variance rather than robust architectural advantages.

S3.2 Probabilistic Verification, Calibration, and Spatiotemporal Block Resampling

Because extreme-precipitation forecasting is intrinsically probabilistic, threshold-dependent binary metrics must be supplemented by proper scoring rules and calibration diagnostics. Across the test partition ($N_{\text{test}} = 528$), the non-parametric Month × Station Climatology establishes a competitive reference Brier score of 0.2224 (Brier Skill Score $\text{BSS}_{\text{uncond}} = +0.0816$ relative to the unconditional training base rate 0.4017, $\text{BS}_{\text{uncond}} = 0.2422$, the Brier score on the test rows of the constant forecast equal to the training

base rate; the test prevalence is 0.411, so this value differs from $\pi(1-\pi) = 0.2403$ at $\pi = 0.4017$). Among trained models, tree ensembles achieve the lowest probability error (Random Forest: BS = 0.2230, $\mathrm{BSS}_{\mathrm{uncond}} = +0.0791$; Gradient Boosting: BS = 0.2236, $\mathrm{BSS}_{\mathrm{uncond}} = +0.0767$), closely matching empirical climatology. Logistic Regression (BS = 0.2258, Expected Calibration Error ECE = 0.0418) and `QKSVM` (BS = 0.2283, ECE = 0.0511) achieve superior probability calibration. In contrast, all hybrid quantum architectures and the standalone LSTM exhibit larger calibration error in the archived single-run evaluation (ECE $\in [0.112, 0.149]$; Brier scores between 0.2352 for `LbL-QNN` and 0.2536 for `LSTM+QNN`), resulting in negative skill scores relative to climatology ($\mathrm{BSS}_{\mathrm{clim}} \in [-0.058, -0.140]$).

Furthermore, because standard DeLong tests assume sample independence that may be compromised by temporal autocorrelation, we conducted a year-block bootstrap analysis ($B = 2{,}000$ resamples of the 19 test calendar years 2008–2026; Künsch, 1989). Resampling entire calendar years across all monitoring stations confirms that Random Forest significantly outperforms `LbL-QNN` (ΔAUC = +0.0151, 95% block CI: $[+0.0007, +0.0281]$, $p = 0.0400$) and Standalone LSTM (ΔAUC = +0.0350, 95% block CI: $[+0.0183, +0.0501]$, $p = 0.0010$). These are single-seed comparisons: the seed-to-seed spread of the same models (Section 4.1) is of the order of the gaps reported. The contrast with the seed-mean comparisons of the main text changes both the training realization and the preprocessing pipeline and therefore does not isolate initialization effects. Meanwhile, Gradient Boosting shows no statistically significant difference from `LbL-QNN` under block resampling (ΔAUC = -0.0010, 95% block CI: $[-0.0335, +0.0315]$, $p = 0.9280$).

S3.3 Feature Importance and Station Indicators

To understand how classical ensembles exploit the feature space, we analyze feature importances across model families. For Random Forest, evaluating Gini feature importance (Mean Decrease in Impurity, MDI) reveals that physically derived thermodynamic features account for 53.8% of total model importance (led by seasonal thermal contrast `choque_clausius` at 6.4%, cumulative evapotranspiration `et0_sum` at 5.3%, and temperature anomaly `t_anomaly` at 3.7%). Antecedent precipitation lags contribute 37.1%, cyclic month encodings account for 5.3%, and geographic station indicator dummies contribute only 3.8%. In contrast, XGBoost gain assigns greater weight to station-specific geographic indicators and cyclic month encodings, followed by thermodynamic proxies.

At face value, the high ranking of geographic indicators in boosted trees might suggest classical ensembles owe their advantage specifically to geographic fixed effects unavailable to the dimension-constrained quantum bottleneck. We test this directly via a station-identity ablation and the capacity-spanning classical control (Section S2.11), and describe the results descriptively: in a single-seed exploratory ablation, removing the station indicators did not degrade classical performance (Section S8), and a classical network restricted to the same 4-dimensional bottleneck had a mean AUC close to that of the quantum circuit. Impurity-based importance does not identify a causal contribution. A plausible reading is that classical ensembles use the full *high-dimensional* feature space—thermodynamic, moisture-lag and seasonal signals—rather than geographic information specifically. This describes what the tree ensembles use; it does not establish that compression causes the gap to the hybrids (Section 4.5).

S3.4 The 15-Seed Replication of the Quantum Tier

To test the stability of the single-run findings, we replicated the training pipeline across 15 independent random-parameter initialization seeds for all gradient-optimized architectures, using a seed-pairing protocol. For the PyTorch-based quantum and deep-learning architectures, seed i is passed to a common `torch.manual_seed` call; we did not verify that this yields identical minibatch order across architectures, so the seed index is treated as a label and not as a pairing device. For cross-paradigm comparisons against the scikit-learn-based classical ensembles (Section S3.5), seed i instead denotes a consistent nominal block index rather than a mechanically shared random stream. Tables S9 and S10 present the resulting distributions for the quantum tier.

An omnibus Friedman test reveals no statistically significant difference in discriminative power among the four quantum architectures ($\chi^2(3) = 2.600$, $p = 0.458$). The effect size is small (Kendall's $W = 0.058$; Demšar, 2006). The seed index is a label rather than a pairing device (see above), so the Friedman test is a descriptive diagnostic of the archived per-seed AUCs, not an inferential test. The single-run AUC of 0.672 for `LbL-QNN` lies above its 15-seed mean of 0.6605. Pairwise Wilcoxon tests indicate that `LbL-QNN` is not statistically superior to any unconstrained counterpart ($p \geq 0.547$ in all cases); notably, unconstrained `LSTM+QNN` achieves the best mean rank of the four (2.200), with `LbL-QNN` second (2.400). The ranks are computed within each seed across the four models, whereas the mean AUC averages the raw values, so a model with a few high-AUC seeds can have the higher mean (`LbL-QNN`, 0.6605 versus 0.6589) without the better mean rank; both differences are negligible against the noise.

To determine whether the absence of a statistically significant difference corresponds to statistical equivalence, we conducted paired Two One-Sided Tests (TOST; Schuirmann, 1987; Lakens, 2017) on seed-level AUC differences. For each model pair, let $d_i = \mathrm{AUC}_{A,i} - \mathrm{AUC}_{B,i}$ denote the paired difference across the $n = 15$ initialization seeds indexed 0–14 (the index is a label, so the paired test is a descriptive diagnostic). We tested $H_{01} : \mu_d \leq -\Delta$ and $H_{02} : \mu_d \geq +\Delta$ using one-sided paired t-tests at $\alpha = 0.05$, with $\Delta = \pm 0.010$ AUC as the reference margin. The numerical TOST criterion was satisfied when both null hypotheses were rejected, equivalently when the paired 90% confidence interval for the mean difference $\bar{d}$ lay entirely within $(-\Delta, +\Delta)$. The margin was defined before conducting the equivalence analysis but was not preregistered; $\Delta = \pm 0.015$ and ± 0.020 are reported as sensitivity analyses. Because the pairing is conventional and the margins lacked a practical justification, no equivalence claim is drawn from these calculations; they do not establish equivalence of generalization to new periods or stations. For `LSTM+QNN` versus `LbL-QNN`, the mean difference is -0.0016 (with standard error $\mathrm{SE}_\Delta = 0.0045$, standard deviation of the paired difference $\mathrm{SD}_\Delta = 0.0173$, and paired seed correlation $r \approx 0.05$), with a paired 90% CI of $[-0.0095, +0.0062]$ contained within the ± 0.010 bounds ($p_{\mathrm{TOST}} = 0.0407$). Under sensitivity margins $\Delta = \pm 0.015$ and ± 0.020, the margin is also satisfied at the wider margins ($p_{\mathrm{TOST}} \leq 0.0028$). These tests concern seed-level optimization only: test-set sampling uncertainty is far larger (standard error of one AUC 0.023–0.027; Section 4.1), and the results do not establish sampling-aware equivalence.

To describe how often one seed outcome exceeds another, we analyzed the complete pairwise cross-seed comparison matrix ($15 \times 15 = 225$ pairs) between `LbL-QNN` and `LSTM+QNN`. Across all 225 realizations, `LbL-QNN` yields a higher test AUC than `LSTM+QNN` in 52.0% of cases (117/225), whereas `LSTM+QNN` is higher in 48.0% of cases (108/225)—a virtually symmetric distribution. Per-seed test probabilities were archived only for seed 0, so no paired DeLong test is made on the 225 pairs; the 15-seed means of the two

models differ by only 0.0016.

The clearest secondary pattern emerging from the quantum tier is lower across-seed variability for the combined layer-wise/heuristic configuration. A depth-matched control for `LbL-QNN` (`LSTM+PI-QNN-3L`) indicates that this variance reduction is not explained by circuit depth alone (Levene's test at matched 3-layer depth, $F_{\text{Levene}} = 7.47$, $p = 0.011$; full analysis in Section S2.6). Because a penalty-free layer-wise control was not evaluated, the separate contributions of the layer-by-layer protocol, the heuristic penalty, and the expanded training budget remain not fully identified by the present design. The multi-seed replication provides no evidence that heuristic thermodynamic regularization lowers the false-alarm rate: the mean FPR of `LbL-QNN` (0.541 ± 0.079) shows no statistically significant difference from the unconstrained `LSTM+QNN` (0.531 ± 0.083; Wilcoxon $p = 0.847$).

Table S9: Global AUC distribution across 15 independent training-seed replications (quantum models only), sorted descending by mean AUC. Standard deviation is the sample standard deviation ($n - 1$ denominator).

Architecture	Mean AUC	Std. Dev.	Minimum	Maximum
LbL-QNN	0.6605	0.0089	0.645	0.681
LSTM+QNN	0.6589	0.0153	0.628	0.681
LSTM+PI-QNN	0.6555	0.0172	0.605	0.673
LSTM+QTL	0.6519	0.0128	0.632	0.679

Table S10: Operational metric distributions (mean ± across-seed sample standard deviation, ddof=1) across 15 training seeds at a default decision threshold of 0.5, for all gradient-optimized and ensemble architectures ($n = 20$ ACP months exceeding P_{60}, $n = 49$ P_{90} months).

Architecture	FPR	Rc-ACP	Rc-P_{90}
Classical Ensembles			
Random Forest	0.383 ± 0.011	0.867 ± 0.041	0.746 ± 0.022
Gradient Boosting	0.310 ± 0.006	0.700 ± 0.000	0.595 ± 0.007
XGBoost	0.408 ± 0.013	0.890 ± 0.047	0.706 ± 0.024
Deep Learning			
Standalone LSTM	0.513 ± 0.066	0.910 ± 0.081	0.782 ± 0.031
Quantum-Classical Hybrid			
LSTM+QNN	0.531 ± 0.083	0.877 ± 0.068	0.785 ± 0.047
LSTM+PI-QNN	0.531 ± 0.048	0.897 ± 0.079	0.796 ± 0.052
LSTM+QTL	0.489 ± 0.075	0.860 ± 0.069	0.762 ± 0.050
LbL-QNN	0.541 ± 0.079	0.867 ± 0.070	0.784 ± 0.045

Rc-ACP is evaluated strictly on the $n = 20$ positive test months ($y = 1$) matching the Autumn Convective Proxy in the canonical pipeline (25 flagged total), distinct from the 32 flagged months (25 positive) under train-only standardization (Section S2.4).

Note on operational variance: Across-seed variance in FPR is substantially higher in gradient-optimized models (LSTMs and quantum hybrids) than in stochastic tree ensembles. Per-architecture across-seed standard deviations are given in Section S9; they are descriptive and suggest that single-run operational comparisons among gradient-based models are unstable.

S3.5 Robustness of the Classical Reference

Extending the 15-seed replication to the classical baselines (Table S5) shows that the *direction* of the classical-over-quantum gap is consistent across seeds, whereas its size is not separated from test-sampling uncertainty by these seed-level data (Section 4.1). Tree-based ensembles exhibit markedly lower across-seed variance ($\sigma_{\text{AUC}} \leq 0.0054$) than any gradient-optimized architecture ($\sigma_{\text{AUC}} \in [0.0089, 0.0171]$).

An omnibus Friedman test across all eight architectures nominally reveals a significant overall ranking effect ($\chi^2(7) = 53.47$, $p < 10^{-6}$; Kendall's $W = 0.509$). However, because Friedman's test formally assumes commensurate, exchangeable blocks across all conditions, treating nominal seed indices across disparate software ecosystems (scikit-learn Mersenne Twister vs. PyTorch/PennyLane PRNG) as paired blocks should be regarded as a heuristic diagnostic rather than an exact inferential test. As a descriptive summary of optimization outcomes, across all 225 cross-framework seed pairs Random Forest (training-row-matched 2,273 rows: 0.6802 ± 0.0035; 2,787 rows: 0.6851 ± 0.0026) has a higher AUC than `LbL-QNN` in 224 of 225 pairs ($\hat{P} = 0.9956$, Cliff's $\delta = 0.9911$, the difference between the probabilities that a draw from one model exceeds a draw from the other and the reverse; Cliff, 1993) and exceeds all other neural and quantum architectures in all pairs (complete matrix in Section S4). These cross-seed summaries describe optimization outcomes on the fixed historical test set. They are not independent replications of generalization performance and do not replace the paired sampling analyses in the main text. However, this optimization-seed separation on a fixed chronological split must not be mistaken for test-set generalization certainty: on the test partition itself, an uncorrected single-run DeLong test comparing Random Forest to `LbL-QNN` yields $p = 0.2124$, and the year-block bootstrap test against empirical climatology yields $p = 0.5420$, indicating that test-set sampling uncertainty ($N = 528$ correlated station-months) remains substantial.

S3.6 Relaxing the Encoding Bottleneck

To test whether the 4D projection bottleneck constrains quantum performance, we evaluated two bandwidth-widened architectures: Amplitude Embedding (16D) and Data Re-uploading (12D across 3 blocks), both projecting *independently and directly* from the full 51-dimensional pre-bottleneck representation—not from an expansion of the pre-existing 4D projection (Section S2.7; Pérez-Salinas et al., 2020), following Table S11.

A descriptive Friedman diagnostic across all six quantum architectures (seed index as a label, not a verified pairing) gives a nominal effect ($\chi^2(5) = 13.057$, $p = 0.0228$; $W = 0.174$), markedly larger than the trivial effect size among the original four ($W = 0.058$; Section S3.4). Both bandwidth-widened architectures achieve the best mean ranks. However, Holm-Bonferroni correction (Holm, 1979) shows that the nominal advantage of Reuploading-3×4D over baseline `LSTM+QNN` ($p = 0.0413$) does not survive multiplicity adjustment ($p_{\text{Holm}} = 0.0826$); and Amplitude-16D shows no statistically significant differentiation from baseline `LSTM+QNN` ($p = 0.1688$). Furthermore, both widened configurations introduce $\sim$ 600–800 classical parameters in the linear projection head, expanding total model capacity. Only Reuploading's advantage over `LbL-QNN` survives ($p_{\text{Holm}} = 0.006$), though that comparison confounds re-uploading with `LbL-QNN`'s distinct training protocol and depth.

The widened variants had lower mean test AUC than Random Forest in these historical runs (primary

matched 2,273 rows: 0.6802 ± 0.0035; 2,787 rows: 0.6851 ± 0.0026; Table S11). Because the comparisons differ in architecture, projection size, depth and training conditions, and sampling-aware paired intervals were not reconstructed, these results do not establish whether widening the encoding interface changes the generalization gap. Notably, while Re-uploading-3×4D exceeds the Standalone LSTM (0.6657 ± 0.0091) and Amplitude-16D approaches it, the bottleneck-matched classical controls (Section S2.11) show no detectable circuit effect over classical linear/MLP projections at a fixed interface, and full-feature XGBoost (0.6575 ± 0.0051) performs comparably to baseline `LSTM+QNN` (0.6589 ± 0.0153). The Re-uploading architecture's improvement is also confounded with an increase in circuit depth relative to the 2-layer `LSTM+QNN` baseline, so it cannot be attributed to encoding bandwidth alone (unlike `LbL-QNN`, where a depth-matched 3-layer control was evaluated, the separate depth-matched control needed to disentangle these factors for re-uploading was not evaluated; see Section 5.3). We therefore report this result descriptively, as evidence that widening the encoding interface is *associated with* improved relative standing among the evaluated quantum architectures, without concluding that bandwidth alone drives the improvement.

Table S11: Global AUC distribution across 15 seeds for the encoding-bandwidth experiments, alongside the original quantum architectures and the classical reference.

Architecture	Mean AUC	Std. Dev.	Mean Rank (of 6)
Reuploading-3×4D	0.6688	0.0085	2.333
Amplitude-16D	0.6643	0.0114	2.933
LSTM+QNN	0.6589	0.0153	3.600
LSTM+PI-QNN	0.6555	0.0171	3.733
LbL-QNN	0.6605	0.0089	3.800
LSTM+QTL	0.6519	0.0128	4.600
Random Forest (matched, 2,273 rows)[*]	*0.6802*	*0.0035*	—
Random Forest (train+val, 2,787 rows)	*0.6851*	*0.0026*	—

[*] Sensitivity baseline trained strictly on 2,273 rows.

S4 Exploratory Analysis of Extreme-Tail and Operational Metrics

S4.1 Small-Sample Binomial Uncertainty in Extreme-Tail Recall

To evaluate model sensitivity in the extreme precipitation tail, the benchmark reports recall metrics conditioned on two diagnostic subsets: months exceeding the regional 90th percentile ($P_{90} = 86.32$ mm, $n = 49$ test months) and months matching the Autumn Convective Proxy (ACP, $n = 25$ test months, of which 20 exceeded P_{60}; throughout this paper, Rc-ACP is defined over the $n = 20$ months with $y = 1$).

Given these small test-partition sample sizes, point estimates of Rc-ACP and Rc-P_{90} carry substantial binomial sampling uncertainty. Table S12 reports 95% Wilson score confidence intervals (Wilson, 1927) for each evaluated architecture in the single representative run. As shown in the table, apparent single-run separations fall well within overlapping confidence intervals. For example, Random Forest's Rc-ACP (0.900, 95% CI $[0.699, 0.972]$) and `LSTM+QNN`'s (0.950, 95% CI $[0.764, 0.991]$) overlap substantially despite a 5-percentage-point difference in the point estimate, underscoring that single-run recall comparisons in the extreme tail reflect test-sample binomial uncertainty independent of optimization variability.

Table S12: Extreme-tail recall with 95% Wilson confidence intervals, single representative run (n = 20 ACP-flagged test months exceeding P_{60} out of 25 flagged total; n = 49 P_{90}-exceeding test months).

Architecture	Rc-ACP [95% CI]	Rc-P_{90} [95% CI]
Climatological Reference		
Month × Station Climatology	0.600 [0.387, 0.781]	0.653 [0.513, 0.771]
Statistical & Ensemble		
Logistic Regression	0.650 [0.433, 0.819]	0.571 [0.433, 0.699]
Random Forest	0.900 [0.699, 0.972]	0.755 [0.619, 0.854]
Gradient Boosting	0.700 [0.481, 0.855]	0.592 [0.452, 0.718]
XGBoost	0.850 [0.640, 0.948]	0.694 [0.555, 0.805]
Deep Learning		
Standalone LSTM	0.950 [0.764, 0.991]	0.816 [0.686, 0.900]
Matched LSTM†	0.950 [0.764, 0.991]	0.776 [0.641, 0.870]
Quantum-Classical Hybrid		
LSTM+QNN	0.950 [0.764, 0.991]	0.796 [0.664, 0.885]
LSTM+PI-QNN	0.950 [0.764, 0.991]	0.755 [0.619, 0.854]
LSTM+QTL	0.850 [0.640, 0.948]	0.837 [0.710, 0.915]
LbL-QNN	0.900 [0.699, 0.972]	0.796 [0.664, 0.885]
QKSVM	0.600 [0.387, 0.781]	0.612 [0.472, 0.736]

†Threshold-recalibrated variant of Standalone LSTM (τ = 0.531) matched to the false-detection rate of `LbL-QNN`.

Rc-ACP is evaluated strictly on the n = 20 positive test months (y = 1) matching the Autumn Convective Proxy in the canonical pipeline (25 flagged total), distinct from the 32 flagged months (25 positive) under train-only standardization (Section S2.4).

S4.2 Deconstructing the Operational Paradox and Threshold Calibration

At the default classification threshold of 0.5, the `Standalone LSTM` achieves high sensitivity to extremes (Rc-ACP = 0.950, 19/20) but suffers from severe overprediction bias (FPR = 0.617). In contrast, `LbL-QNN` appears more conservative (FPR = 0.495, Rc-ACP = 0.900, 18/20), suggesting an apparent operational advantage for the hybrid architecture (Table S8).

However, this apparent advantage is sensitive to the selected operating threshold. When the classical LSTM is adjusted to operate at an approximately matched FPR of ≈50.2% (versus 49.5%) (by raising its decision threshold to τ = 0.531, denoted `Matched LSTM`; τ was chosen on the test set to match the FPR, so this is a retrospective exploratory comparison and not an evaluation of a decision rule calibrated on validation data), its ACP recall remains at 0.950 (19/20). Paired bootstrap analysis (n = 2,000 resamples) reveals no statistically significant operational difference between the two models across the operational frontier (all 95% confidence intervals span zero).

Within-run uncertainty estimation can mislead architectural comparison: the bootstrap confidence intervals and DeLong p-values from the single historical realization nominally favor `LbL-QNN` over both unconstrained `LSTM+QNN` (p = 0.0070) and `Standalone LSTM` (p = 0.0263). Both are conditioned on one initialization draw, and the 15-seed means (Section S3.4) are 0.6605 and 0.6589 for the first two.

S5 Training Hyperparameters of the Historical Experiments

The Core experiment uses the configuration of Table S3, which differs in places from this historical configuration (for example the LSTM dropout). Table S13 reports the full hyperparameter configuration for all evaluated architectures, to support exact reproducibility.

Table S13: Training hyperparameters for all evaluated architectures. All gradient-optimized models use identical early-stopping criteria (patience = 6 epochs monitoring BCE validation loss) unless noted in *Other*.

Architecture	Optimizer	LR	Batch	Epochs	Other
Logistic Regression	L-BFGS	—	—	—	L2 penalty, $C = 0.1$
Random Forest	—	—	—	—	300 trees, max_depth=8
Gradient Boosting	—	0.05‡	—	—	200 est., subsample=1.0, max_features=None
XGBoost	—	0.05‡	—	—	500 est., depth=4, subsample=0.8, colsample_bytree=0.8, scale_pos_weight = $n_{\text{neg}}/n_{\text{pos}}$
Standalone LSTM	AdamW	0.005	16	25	16 hidden units/direction (32D concat.), dropout=0.2
LSTM+QNN	AdamW	0.005	16	25	4 qubits, 2 layers
LSTM+PI-QNN	AdamW	0.005	16	25	$\lambda_{cc} = 0.2, \lambda_{\text{hMSE}} = 0.1$
LSTM+QTL	AdamW	0.002	8	15	frozen LSTM, patience=4
LbL-QNN	AdamW	var.*	16	55**	freeze: 10 eps/layer, pat=7
LSTM+MLP-4D (Linear-20)	AdamW	0.005	16	25	Linear(4→4), 20 params, exact structural parity
LSTM+MLP-4D (LeakyReLU-28)	AdamW	0.005	16	25	Linear(4→2) + LeakyReLU + Linear(2→2) + LeakyReLU + Linear(2→4), 28 params, He init
LSTM+MLP-4D (GELU-180)	AdamW	0.005	16	25	Linear(4→16) +GELU + LN(16) + Linear(16→4), 180 params

* `LbL-QNN` uses 5×10^{-3} during layer-wise freezing and 5×10^{-4} during global fine-tuning.

** Total epoch budget: 10 epochs per frozen layer (3 layers) plus 25 epochs of global fine-tuning.

‡ For tree ensembles, "LR" denotes the boosting shrinkage parameter, not a gradient-descent step size.

The classical ensemble hyperparameters and the quantum penalty weights (λ_{cc}, λ_{hMSE}) were manually specified; a reproducible tuning history is unavailable. Matched human tuning effort does not equate to matched compute budgets (Bowles et al., 2024), and equal tuning effort across model families should not be assumed. The variance differences documented in this paper were not isolated from tuning effort.

S6 Inference-Time Noise Robustness (Supplementary)

As a preliminary check, we evaluate the noiseless-trained parameters under simulated depolarizing gate noise ($p = 0.01$) at inference time only; this does not address noise-induced barren plateaus during training (Wang et al., 2021) and should not be read as evidence of hardware viability. The resulting degradation is reported in Table S14.

Table S14: Predictive metric degradation under simulated depolarizing gate noise ($p = 0.01$, 15-seed Monte Carlo over noise draws on fixed weights).

Architecture	Environment	Accuracy (%)	F1-Score	Rc-ACP	FPR
LSTM+QNN	Noiseless	58.0	0.599	0.950	0.550
	Noisy ($p = 0.01$)	57.8	0.595 ± 0.006	0.933 ± 0.024	0.545 ± 0.009
LbL-QNN	Noiseless	59.1	0.589	0.900	0.495
	Noisy ($p = 0.01$)	59.3	0.593 ± 0.007	0.887 ± 0.039	0.495 ± 0.017

Rc-ACP evaluated on the canonical $n = 20$ positive test months (25 flagged total; distinct from the 32 flagged and 25 positive months under train-only standardization).

Both models display notable structural robustness to inference-time gate noise. The layer-wise model (`LbL-QNN`) closely preserves its operational FPR under noise (0.495 ± 0.017, matching its noiseless value to three decimal places), though with non-negligible dispersion in Rc-ACP. We confirm this evaluation strictly measures inference-time degradation on parameters obtained from noiseless training: forward passes are computed under `torch.no_grad()` on weights loaded from the clean-trained state dictionary, with no optimizer or gradient step present. The 15-seed loop characterizes the stochasticity of the noise channel and finite-shot sampling (shots = 1024), not training-seed variance. Noise-aware training—optimizing directly under the noise model rather than evaluating post hoc—would provide substantially stronger evidence; we did not perform this owing to the computational cost of a full 15-seed replication under simulated noisy gradients; we consider it the most consequential open extension of the noise analysis presented here. This evaluation is currently restricted to `LSTM+QNN` and `LbL-QNN` (the unconstrained and physics-regularized endpoints); extending it to `LSTM+PI-QNN` and `LSTM+QTL` is a straightforward extension for future work.

S7 Heuristic Thermodynamic Regularization, Latent Coordinates, and Sensitivity Sweep

S7.1 Formulation of Heuristic Thermodynamic Loss Penalties

To explore whether domain-motivated soft constraints alter optimization behavior, we constructed two heuristic regularizers combining binary cross-entropy ($\mathcal{L}_{\text{BCE}}$) with physical penalties:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \alpha(e)\,[\lambda_{\text{cc}}\mathcal{L}_{\text{CC}} + \lambda_{\text{hMSE}}\mathcal{L}_{\text{hMSE}}] \tag{S2}$$

where $\alpha(e) = \min(1.0, e/(0.6 \times E_{\text{total}}))$ represents a linear annealing schedule across training epochs e.

1. Clausius–Clapeyron-Inspired Gating Penalty ($\mathcal{L}_{\text{CC}}$): Defines a temperature-dependent soft gate:

$$C_{\text{CC}} = \exp(0.07 \cdot T_{\text{anomaly}} \cdot 10.0) \tag{S3}$$

loosely motivated by the exponential scaling of saturation water vapour pressure with temperature ($\approx 7\%/\text{K}$). A ReLU-gated penalty activates if the model predicts high precipitation probability ($P_{\text{rain}} > 0.6$) when

this thermal gate is restricted:

$$\mathcal{L}_{\text{CC}} = \frac{1}{N}\sum_{i=1}^{N}\left[\max(0, P_{\text{rain},i} - 0.6) \times \max(0, 1.0 - C_{\text{CC},i})\right] \tag{S4}$$

Dimensional and Physical Caveat: We emphasize that the scale factor 0.07 originates as a per-kelvin physical sensitivity from the Clausius–Clapeyron relation, but in the implementation it is multiplied by 10.0 and evaluated directly on a dimensionless, standardized temperature anomaly z-score (T_{anomaly}). Furthermore, the 3-month lagged thermal difference (`choque_clausius`) reflects seasonal transition cooling rather than saturation thermodynamics. Consequently, C_{CC} is an empirical phenomenological regularizer rather than a dimensionally consistent thermodynamic equation.

2. Heuristic Latent-Coordinate Balance Penalty ($\mathcal{L}_{\text{hMSE}}$): Encourages the readout expectation values nominally associated with sensible heat (q_0) and latent heat (q_1) to balance the coordinate nominally associated with convective release (q_3):

$$\mathcal{L}_{\text{hMSE}} = \frac{1}{N}\sum_{i=1}^{N}\left[\max(0, q_{0,i}) + \max(0, q_{1,i}) - \max(0, q_{3,i}) - 0.5\right]^2 \tag{S5}$$

where $q_j = \langle 0|U^\dagger(\theta, x)\sigma_z^{(j)}U(\theta, x)|0\rangle$ denotes the expectation value of Pauli-Z on qubit j. We use the notation $\mathcal{L}_{\text{hMSE}}$ to distinguish this heuristic coordinate penalty from standard regression Mean Squared Error.

The internal constants (0.07, 10.0, 0.6, 0.5) and outer penalty weights ($\lambda_{\text{cc}} = 0.2$, $\lambda_{\text{hMSE}} = 0.1$) were set by hand, with no tuning protocol recorded, based on qualitative heuristic reasoning and held fixed during main benchmark training.

Physics-informed learning embeds domain constraints in the loss (Raissi et al., 2019; Karniadakis et al., 2021), including in quantum models (Jaderberg et al., 2024); the penalties used here are a loose, heuristic version of that idea. Exact architectural symmetries, such as equivariant quantum neural networks (Meyer et al., 2023; Nguyen et al., 2024), are a more principled direction for physics-informed QML.

S7.2 Latent Readout Coordinate Interpretations

Table S15 details the descriptive interpretations heuristically assigned to the four Pauli-Z readout expectations. These labels are descriptive post-hoc conventions and do not correspond to physical thermodynamic state variables or conservation laws.

Table S15: Post-hoc interpretation of the four measured latent coordinates. The labels are descriptive only and do not imply thermodynamic units, state identification, or physically enforced energy conservation.

Qubit	Latent Coordinate	Heuristic interpretation
Q_0	Thermal Energy Coordinate (q_0)	Thermal-related latent coordinate
Q_1	Moisture Coordinate (q_1)	Moisture-related latent coordinate ($L_v q$)
Q_2	Dynamic Coordinate (q_2)	Dynamic-related latent coordinate
Q_3	Convective Release Coordinate (q_3)	Convective-related latent coordinate

S7.3 Exploratory Physical Penalty Weight (λ_{cc}) Sensitivity Logging

A sensitivity sweep over λ_{cc} was only partially recorded in the archived artifacts and cannot be reconstructed; no results from it are reported.

S8 Spatial Memorization and Label Leakage Ablations

To verify that the classical performance protocol is not driven by spatial memorization or label leakage, we conducted two targeted ablations on the XGBoost architecture.

1. Spatial Memorization: As an exploratory single-run check (not subject to our 15-seed replication standard), we retrained XGBoost with the four station indicator variables (`ciudad_*`) removed. Consistent with the hypothesis that geographic dummies do not substitute for genuine dynamical signal, single-run AUC did not degrade (0.6495 $\rightarrow$ 0.6541, ΔAUC = +0.0046), and thermodynamic features rose in the importance ranking. However, because this check was conducted on a single initialization seed, it is reported strictly as descriptive preliminary evidence rather than a confirmed architectural invariant.

2. Label Leakage: The diagnostic *acp_flag* relies partly on an anomalous wind-gust index (*ia_racha*), also present among the training features. Removing *ia_racha* produced no detected change in the 15-seed metrics (mean AUC: 0.6575 $\pm$ 0.0051 $\rightarrow$ 0.6585 $\pm$ 0.0048, $p = 0.890$; mean Rc-ACP: 0.890 $\pm$ 0.047 $\rightarrow$ 0.900 $\pm$ 0.054, $p = 0.587$). The slight, non-significant *increase* in Rc-ACP is the opposite of what a label shortcut through this variable would predict, but the ablation does not identify the source of the ACP sensitivity. The antecedent gust index used as a predictor and the target-month gust index used to define the ACP subset are different quantities; the absence of temporal leakage rests on that timeline, which must be checked against the feature-construction code.

S9 Descriptive Across-Seed Variability of Operational Metrics

Here we report the across-seed variability of the false-positive rate (FPR) per architecture. Gradient-optimized models have FPR standard deviations 3.5 to 7.4 times those of the stochastic tree ensembles (Random Forest $\sigma = 0.011$; XGBoost $\sigma = 0.013$); these are ratios of standard deviations, not of variances. Pooled variance tests mix between-architecture differences in means with within-architecture seed variability and are therefore not used as evidence. Gradient Boosting was configured without row or feature subsampling (`subsample=1.0`, `max_features=None`); its small non-zero standard deviation ($\sigma_{FPR} \approx 0.006$) was not investigated. These descriptive results do not imply that optimization variability is larger for all gradient-based models or that the ratios generalize beyond this benchmark.

S10 Circuit and Computational Cost Analysis

This appendix provides the complete software environment specification (Table S16), native gate decomposition (Table S17), hybrid parameter accounting (Table S18), dimensionality reduction costs (Table S19), shot-based execution projections (Table S20), and total computational cost summary (Table S21) for all evaluated quantum architectures. All gate counts were verified via PennyLane 0.45 `QuantumTape` decomposition to native {RY, RZ, RX, CNOT} gates.

S10.1 Software Environment and Hardware

Table S16: Software environment and simulation hardware.

Component	Version/Specification
PennyLane	0.45.0
PyTorch	2.11.0+cpu
Python	3.12.0 (MSC v.1935 64-bit AMD64)
NumPy	2.0.2
Qiskit Aer	≥0.17 (NISQ noise simulation)
Simulator	`default.qubit` (PennyLane statevector)
Differentiation	`diff_method='backprop'` (analytical, through statevector)
Hardware	CPU-only (`cpuonly` in environment.yml, `+cpu` in PyTorch build)
NISQ noise sim	`qiskit.aer` with depolarizing noise model, `shots=1024`

S10.2 Circuit Architecture: Decomposed Gate Counts

Table S17: Circuit decomposition to native gates for all evaluated quantum architectures.

Circuit	Qubits	Depth	Total Gates	Single-Qubit	CNOTs	Quantum Params
LSTM+QNN / PI-QNN (4q, 2L)	4	18	48	40	8	24
LbL-QNN (4q, 3L)	4	27	72	60	12	36
QTL (4q, 2L)	4	13	36	28	8	24
QKSVM kernel (4q)	4	2	8	8	0	0 (data only)
Encoding-bandwidth ablation circuits						
Amplitude Embedding (4q, 2L)	4	27	50*	28	22	24
Re-uploading-3×4D (4q, 3L)	4	27	72	60	12	36

*Amplitude state-preparation gate count depends on decomposition scheme (50 total gates and depth 27 under PennyLane default Möttönen state preparation). Circuit depths are compiled on the native gate set $\{RY, RZ, CNOT\}$ using PennyLane's DAG critical-path analysis.

Gate decomposition detail (LSTM+QNN, 4q-2L): Per layer, the encoding block applies $4\times\text{RY}+4\times\text{RZ} = 8$ single-qubit gates (data embedding via RY and RZ rotations). The `StronglyEntanglingLayers` block applies $4 \times \text{Rot}(\theta, \phi, \omega) \rightarrow 4 \times \text{RY} + 8 \times \text{RZ} + 4 \times \text{CNOT} = 16$ gates. Totals for 2 layers: 8 RY(data) + 8 RZ(data) + 8 RY(param) + 16 RZ(param) + 8 CNOT = **48 gates** (16 RY, 24 RZ, 8 CNOT). LbL-QNN adds a third block, totaling **72 gates** (24 RY, 36 RZ, 12 CNOT). QTL uses **36 gates** (12 RY, 16 RZ, 8 CNOT). Re-uploading-3×4D uses 3 sequential encoding and entangling blocks, totaling **72 gates** (24 RY, 36 RZ, 12 CNOT). Compiled critical-path circuit depths on the native $\{RY, RZ, CNOT\}$ basis are 18 for LSTM+QNN / PI-QNN, 27 for LbL-QNN, 13 for QTL, 2 for QKSVM, 27 for Amplitude Embedding, and 27 for Re-uploading-3×4D.

S10.3 Hybrid Model Parameter Accounting

Key observation: In LSTM+QNN, only 24/2,613 = **0.92% of trainable parameters** are quantum. The circuit adds 48 gates on a 16-dimensional state vector atop a 2,589-parameter classical backbone (2,368-parameter LSTM encoder + 208-parameter projection layer + 13-parameter normalization and head). For QTL, the entire PureLSTM encoder (10,103 params) is frozen; only the adapter network (580 params) + head/norm (13 params) + quantum circuit weights (24 params) are trainable = 617 trainable

Table S18: Complete parameter accounting for hybrid quantum-classical models. `in_dim` = 35 for 4 stations (12 precipitation lags + 17 physics features + 2 seasonal + 4 station dummies).

Model	Total Params	Trainable Params	Quantum (W)	Classical
LSTM+QNN (4q, 2L)	2,613	2,613	24	2,589
LSTM+PI-QNN (4q, 2L)	2,613	2,613	24	2,589
LbL-QNN (4q, 3L)	2,625	2,625	36	2,589
QTL (4q, 2L)	10,720	**617**	24	593 (adapter+head)
PureLSTM (baseline)	10,103	10,103	—	10,103
Amplitude Embedding (4q, 2L)	3,237	3,237	24	3,213
Re-uploading-3×4D (4q, 3L)	3,041	3,041	36	3,005
Classical bottleneck controls				
Linear parity control	2,609	2,609	0	2,609
LeakyMLP parity control	2,617	2,617	0	2,617
GELU MLP control	2,769	2,769	0	2,769

parameters (3.89% quantum).

S10.4 Dimensionality Reduction Cost

The full input dimensionality is 35 (with 4 stations). Hybrid models concatenate the 16-dimensional bidirectional LSTM representation with the 35 raw features, yielding a 51-dimensional intermediate representation. The circuit operates on only 4 qubits. There is a classical projection from 51 → 4 hidden inside the models.

Table S19: Dimensionality reduction pipeline and associated classical cost.

Model	Reduction Method	From → To	Params in Reduction
LSTM+QNN / PI-QNN	`nn.Linear(16 + in_dim, n_q)`	51 → 4	51×4 + 4 = **208**
QTL	`nn.Linear(67, 8)` → `nn.Linear(8, 4)`	67 → 8 → 4	67×8+8 + 8×4+4 = **580**
QKSVM	PCA(`n_components=4`) + MinMaxScaler	35 → 4	0 trainable (eigendecomposition: $O(d^2N + d^3)$)
Amplitude	`nn.Linear(16 + in_dim, 2^n_q)`	51 → 16	51×16 + 16 = **832**
Re-uploading (3 blocks)	`nn.Linear(16 + in_dim, 12)`	51 → 12	12×51 + 12 = **624**

The LSTM encoder before the projection compresses the 12 precipitation lags into a 16-dim hidden state (bidirectional LSTM, 2 layers, hidden=8), adding:

- LSTM: `nn.LSTM(1, 8, 2, bidirectional=True)` → **2,368 parameters**
- The projection sees: **16 (LSTM hidden) + 35 (in_dim) → 51**

Implication: Including the full classical preprocessing (LSTM encoder + projection) means that most trainable parameters are classical. The circuit adds 48 gates on a 16-dimensional state vector atop a 2,589-parameter classical backbone; the parameter count alone does not quantify the functional contribution of the circuit.

S10.5 Circuit Evaluations and Computational Cost

S10.5.1 Statevector mode (`diff_method='backprop'`)

- **Forward pass:** 1 circuit evaluation per sample (the `for xi in qi` loop in `forward()`)

- **Backward pass:** no additional circuit executions (the gradient is obtained by backpropagation through the statevector, which has its own cost)
- **Per epoch:** N_{train} circuit evaluations (forward only)
- **Per training run (25 epochs):** $\leq 25 \times N_{\text{train}}$

State vector size: $2^4 = 16$ complex amplitudes. Memory per state: 16×16 bytes (complex128) = 256 bytes for the basic statevector (excluding autograd storage). Backprop gradient cost: $O(2^n \times \text{depth})$ per sample. The 4-qubit statevector is trivially small; the computational bottleneck is the Python-level `for xi in qi` loop (sequential, no batching across samples in the quantum part).

S10.5.2 Estimated Cost With Shots (Hardware / Parameter-Shift)

If running on real quantum hardware (parameter-shift rule instead of backprop):

Table S20: Circuit evaluations per sample under parameter-shift rule with $N_{\text{shots}} = 1024$.

Component	LSTM+QNN (24 params)	LbL-QNN (36 params)
Forward	1 circuit × 1024 shots	1 circuit × 1024 shots
Backward (parameter-shift)	$2 \times 24 = 48$ circuits × 1024 shots	$2 \times 36 = 72$ circuits × 1024 shots
Total per sample	**49 circuits × 1024 shots**	**73 circuits × 1024 shots**
Total shots per sample	**50,176**	**74,752**

The table counts parameter-shift evaluations for the variational circuit weights only. It excludes the input-angle derivatives required for end-to-end training, execution overhead and hardware latency, and is not a measured runtime ratio relative to statevector backpropagation. Finite sampling adds shot noise.

S10.6 Summary: Gate, Parameter and Partial Execution-Count Profile

Table S21: Gate, parameter and partial execution-count summary for the quantum architectures (excludes runtime, input-angle derivatives and hardware overheads).

Metric	LSTM+QNN	PI-QNN	LbL-QNN	QTL	QKSVM
Qubits	4	4	4	4	4
Circuit depth	18	18	27	13	2
Single-qubit gates	40	40	60	28	8
CNOTs	8	8	12	8	0
Total gates	48	48	72	36	8
Quantum params	24	24	36	24	0 (fixed map)
Total trainable params	2,613	2,613	2,625	617	N/A (SVM)
Classical params	2,589	2,589	2,589	593	PCA+SVM
% quantum params	**0.92%**	**0.92%**	**1.37%**	**3.89%**	**Fixed map***
Statevector cost	$O(16 \times \text{depth})$	$O(16 \times \text{depth})$	$O(16 \times \text{depth})$	$O(16 \times \text{depth})$	$O(16)$
Shots cost ($N = 1024$)	49×1024/sample	49×1024/sample	73×1024/sample	49×1024/sample	1×1024/kernel
Diff method	backprop	backprop	backprop	backprop	N/A
Simulator	default.qubit	default.qubit	default.qubit	default.qubit	default.qubit
Hardware	CPU (statevector)	CPU (statevector)	CPU (statevector)	CPU (statevector)	CPU (statevector)
Max epochs	25	25	55 (30+25)	15	N/A
Dim. reduction	51→4 (linear)	51→4 (linear)	51→4 (linear)	67→4 (2-layer)	35→4 (PCA)

*QKSVM utilizes a fixed 4-qubit quantum feature map without variational parameters; optimization consists entirely of classical quadratic programming in the dual space.

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